MINIMUM GEODESIC OCCUPATION OF INDEPENDENT RANDOM CHECKERBOARDS CONVERGES TO THEIR DENSITY

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Editorial note. Complete original mathematical body, all author probability/geodesic arguments and10 bibliography entries retained with7 original figures. Original fixed-time/Bernoulli density hypotheses unchanged; convergence asserted in probability as in source. Native EPS-converted original PDF Form is used by changing only its filename extension in the rendering copy.

Editorial note. Two external PDF-TeX overlays are flattened as complete original printed vector diagrams with all author labels, excluding captions retained in source. Original include widths unchanged; rendering-copy replacements preserve line count.

Allowed standard prerequisites (editorial).

- Standard functional analysis, measure theory and the author-cited classical theorems. Standard prior results stay cited; complete source-local proofs and definitions retained.

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1. Introduction and main results

Let (M, g) be a closed connected Riemannian manifold. We denote by Γ the set of geodesic rays, that is, the set of projections onto M of Riemannian geodesic curves in the co-sphere bundle S^*M. Given any $T > 0$ and any Lebesgue measurable subset ω of M, we define

$$\ell^T(\omega) = \inf_{\gamma\in\Gamma} \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\,dt. \tag{1}$$

Here, χ_ω is the characteristic function of ω, defined by $\chi_\omega(x) = 1$ if $x \in \omega$ and $\chi_\omega(x) = 0$ if $x \in M \setminus \omega$. The real number $\ell^T(\omega) \in [0, 1]$ is the smallest average time over $[0, T]$ spent by all geodesic rays in ω. This quantity appears naturally when studying observability properties for the wave equation on M with ω as an observation subset.

In this article we establish two properties of the functional ℓ^T, one is geometric and the other is probabilistic. Let us describe them in few words.

The first geometric property is on the maximal discrepancy of ℓ^T when taking the closure. We may have $\ell^T(\mathring{\omega}) < \ell^T(\overline{\omega})$ whenever there exist rays grazing ω and the discrepancy between both quantities may be equal to 1 for some subsets ω. We prove that, if the metric g is C^2 and if ω satisfies a slight regularity assumption, then $\ell^T(\overline{\omega}) \leq \frac{1}{2}\left(\ell^T(\mathring{\omega}) + 1\right)$. We also show that our assumptions are essentially sharp; in particular, surprisingly the result is wrong if the metric g is not C^2. As a consequence, if ω is regular enough and if $\ell^T(\overline{\omega}) > 1/2$ then the Geometric Control Condition is satisfied and thus the wave equation is observable on ω in time T.

The second property is of probabilistic nature. We take $M = \mathbb{T}^2$, the flat two-dimensional torus, and we consider a regular grid on it, a regular checkerboard made of n^2 square white cells. We construct random subsets ω_ε^n by darkening each cell in this grid with a probability ε. We prove that the random law $\ell^T(\omega_\varepsilon^n)$

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converges in probability to ε as $n \to +\infty$. As a consequence, if n is large enough then the Geometric Control Condition is satisfied almost surely and thus the wave equation is observable on ω_ε^n in time T.

Observability and Geometric Control Condition. The condition

$$\ell^T(\omega) > 0$$

means that all geodesic rays, propagating in M, meet ω within time T. This condition, usually called *Geometric Control Condition* (in short, GCC), is related to observability properties for the wave equation

$$\partial_{tt} y - \triangle_g y = 0 \qquad \text{in } (0,T)\times M, \tag{2}$$

where $\triangle_g$ is the Laplace-Beltrami operator on M for the metric g. More precisely, denoting by dx_g the canonical Riemannian volume, we define the observability constant $C_T(\omega) \geq 0$ as the largest possible nonnegative constant C such that the inequality

$$\int_0^T \int_\omega |y(t,x)|^2 \, dx_g \, dt \geq C\Big(\|y(0,\cdot)\|^2_{L^2(M)} + \|\partial_t y(0,x)\|^2_{H^{-1}(M)} \Big) \tag{3}$$

is satisfied for any solution y of (2), that is,

$$C_T(\omega) = \inf\left\{ \int_0^T \int_\omega |y(t,x)|^2 \, dx_g \, dt \mid \|(y(0,\cdot),\partial_t y(0,\cdot))\|_{L^2(M)\times H^{-1}(M)} = 1 \right\}.$$

When $C_T(\omega) > 0$, the wave equation (2) is said to be *observable* on ω in time T, and when $C_T(\omega) = 0$ we say that observability does not hold for (ω, T).

It has been proved in [1, 9] that, for ω open, observability holds if the pair (ω, T) satisfies GCC, i.e.if $\ell^T(\omega) > 0$. In other words, if ω is open and satisfies $\ell^T(\omega) > 0$ then the wave equation (2) is observable on ω in time T.

The converse is not true: GCC is not a necessary condition for observability. It is shown in [8] that, if $M = \mathbb{S}^2$ (the unit sphere in $\mathbb{R}^3$ endowed with the restriction of the Euclidean structure), if ω is the open Northern hemisphere, then $\ell^T(\omega) = 0$ for every $T > 0$, and however one has $C_T(\omega) > 0$ for every $T > \pi$. The latter fact is established by an explicit computation exploiting symmetries of solutions. This failure of the functional ℓ^T to capture the observability property is due, here, to the existence of a very particular geodesic ray which is *grazing* the open set ω, namely, the equator. In this example, considering the closure $\overline{\omega}$ of ω, it is interesting to observe that $\ell^T(\overline{\omega}) = 0$ for every $T \leq \pi$ (take a geodesic ray contained in the closed Southern hemisphere) and $\ell^T(\overline{\omega}) > 0$ for every $T > \pi$, with $\ell^T(\overline{\omega}) = \frac{1}{2}$ when $T \geq 2\pi$. The latter equality is in contrast with $\ell^T(\omega) = 0$: there is thus a discrepancy $1/2$ in the value of ℓ^T for $T \geq 2\pi$ when taking the closure of ω. In this specific case, this discrepancy is caused by the equator, which is a geodesic ray grazing the open subset ω.

Our first main result below shows that $1/2$ is actually the maximal possible discrepancy.

1.1. **A geometric result on the maximal discrepancy of ℓ^T.** In general, one can always find subsets ω for which the difference $\ell^T(\overline{\omega}) - \ell^T(\mathring{\omega})$ is arbitrary close to 1. Surprisingly, under slight regularity assumptions, this maximal discrepancy is $1/2$ only.

Theorem 1. *Let $T > 0$ be arbitrary and let ω be a measurable subset of M. We make the following assumptions:*

(i) *The metric g is at least of class C^2.*

(ii) *ω is an embedded C^1 submanifold of M with boundary if $\dim M \geq 3$ and is piecewise C^1 if $\dim M = 2$.*

Then

$$\ell^T(\overline{\omega}) \leq \frac{1}{2}\big(\ell^T(\mathring{\omega}) + 1\big). \tag{4}$$

We give more details and a number of comments on this theorem in Section 2. At the opposite, as an obvious remark, if there is no geodesic ray grazing ω then $\ell^T(\overline{\omega}) = \ell^T(\mathring{\omega})$. Here and throughout the paper, we say that a geodesic ray γ is grazing ω if $\int_0^T \chi_{\partial\omega}(\gamma(t))\,dt > 0$, where $\partial\omega = \overline{\omega} \setminus \mathring{\omega}$.

In more general, the existence of grazing rays adds a serious difficulty to the analysis of observability (see [1]). It is noticeable that, if one replaces the characteristic function χ_ω of ω by a continuous function a, in the integral at the left-hand side of (3) (i.e.$\int_0^T \int_M a(x)|y(t,x)|^2\,dx_g\,dt$) as well as in the definition (1) of the functional ℓ^T, this difficulty disappears and the condition $\ell^T(a) > 0$ becomes a necessary and sufficient condition for observability of (2) on ω in time T (see [4]).

By the way, for completeness, we provide in the appendix some semi-continuity properties of the functional ℓ^T, which may be of interest for other purposes.

The issue of the observability on a general measurable subset $\omega \subset M$ has remained widely open for a long time. Recent advances have been made, which we can summarize as follows. It has been established in

[7] that observability on a measurable subset ω in time T is satisfied if and only if $\alpha_T(\omega) > 0$. The quantity $\alpha_T(\omega)$, defined in [7] as the limit of high-frequency observability constants, is however not easy to compute and we have, in general, the inequality $\ell^T(\mathring{\omega}) \leq \alpha_T(\omega) \leq \ell^T(\overline{\omega})$. In particular, the condition $\ell^T(\omega) > 0$ becomes a necessary and sufficient condition for observability as soon as there are no geodesic rays grazing ω. It has also been shown in [7] that $\lim_{T\to+\infty} {}^{C_T(\omega)}/_T$ is the minimum of two quantities, one of them being $\ell^T(\overline{\omega})$ and the other being of a spectral nature.

We have the following corollary of Theorem 1, using the fact that, since $\mathring{\omega}$ is open, the condition $\ell^T(\mathring{\omega}) > 0$ implies observability for $(\mathring{\omega}, T)$, and thus $C_T(\omega) \geq C_T(\mathring{\omega}) > 0$.

Corollary 1. *Under the assumptions of Theorem 1, if $\ell^T(\overline{\omega}) > 1/2$ then $\ell^T(\mathring{\omega}) > 0$ and thus the wave equation (2) is observable on ω in time T, i.e.$C_T(\omega) > 0$.*

Note that Corollary 1 does not apply to the (limit) case where $M = \mathbb{S}^2$ and ω is the open Northern hemisphere. It does neither apply to the case where M is the two-dimensional torus and ω is a half-covering open checkerboard on it, as in [3, 5] (see next section). Indeed, in these two cases, we have $\ell^T(\omega) = 0$ for every $T > 0$ but $C_T(\omega) > 0$ (i.e.we have observability) for T large enough. This is due to the fact that trapped rays are the weak limit of Gaussian beams that oscillate *on both sides* of the limit ray, spreading on one side and on the other a sufficient amount of energy so that indeed observability holds true. In full generality, having information on the way that semi-classical measures, supported on a grazing ray, can be approached by high-frequency wave packets such as Gaussian beams, is a difficult question. In the case of the sphere, symmetry arguments give the answer (see [8]). In the case of the torus, a much more involved analysis is required, based on second microlocalization arguments (see [3, 5]).

Anyway, Corollary 1 can as well be applied for instance to any kind of checkerboard domain ω on the two-dimensional torus, as soon as the measure of ω is large enough so that $\ell^T(\overline{\omega}) > 1/2$.

Since the case of checkerboards (in dimension two) is interesting and challenging, following a question by Nicolas Burq, in the next section we investigate the case of random checkerboards on the flat torus and we establish our second main result.

1.2. **A probabilistic result for random checkerboards on the flat torus.** In this section, we take $M = \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ (flat torus) which is identified to the square $[0,1]^2$, class of equivalence of $\mathbb{R}^2$ under the identifications $(x, y) \sim (x + 1, y) \sim (x, y + 1)$, inheriting of the Euclidean metric. Given any subset A of M, we denote by $|A|$ the (two-dimensional) Lebesgue measure of A.

We consider a regular grid $\mathcal{G}^n = (c_{ij}^n)_{1\leq i,j\leq n}$ in the square, like a checkerboard, made of $n \times n$ closed squares:

$$[0,1]^2 = \bigcup_{i,j=1}^{n} c_{ij}^n \quad \text{with} \quad c_{ij}^n = [(i-1)/n, i/n] \times [(j-1)/n, j/n] \quad \forall (i,j) \in \{1,\dots,n\}^2.$$

Defining $c_{i'j'}$ in the same way for all $(i', j') \in \mathbb{Z}^2$, we identify the square $c_{i'j'}$ to the square c_{ij}^n of the above grid with $(i, j) \in \{1, \dots, n\}^2$ such that $i = i' \mod n$ and $j = j' \mod n$.

Construction of random checkerboards. Let $\varepsilon \in [0, 1]$ be arbitrary. Considering that all squares in the grid are initially white, we construct a random checkerboard by randomly darkening some squares in the checkerboard as follows: for every $(i, j) \in \{1, \dots, n\}^2$, we darken the square c_{ij}^n of the grid with a probability ε. All choices are assumed to be mutually independent. In other words, we make a selection of squares (that are paint in black) in the grid by considering n^2 independent Bernoulli random variables denoted $(X_{ij}^n)_{1\leq i,j\leq n}$, each of them with parameter ε. The total number of black squares follows therefore the binomial law $B(n^2, \varepsilon)$.

We denote by ω_ε^n the resulting closed subset of $[0,1]^2$ that is the union of all (closed) black squares (see Figure 1).

Given any fixed $T > 0$ and $\varepsilon \in (0, 1]$, our objective is to understand how well the random set ω_ε^n is able to capture all geodesic rays propagating in $M \simeq [0,1]^2$, in finite time T. In other words, we want to study the random variable $\ell^T(\omega_\varepsilon^n)$. Of course, the random variable $|\omega_\varepsilon^n|$ follows the law ${}^{(1}/_{n^2)}B(n^2, \varepsilon)$ and thus its expectation is equal to ε, and so, when ε is small, ω_ε^n covers only a small area in $[0,1]^2$. And yet, our second main result below shows that, for n large, almost all such random sets meet all geodesic rays within time T.

Theorem 2. *Given any $T > 0$ and any $\varepsilon \in [0, 1]$, the random variable $\ell^T(\omega_\varepsilon^n)$ converges in probability to ε as $n \to +\infty$, i.e.*

$$\lim_{n\to+\infty} \mathbb{P}\left(|\ell^T(\omega_\varepsilon^n) - \varepsilon| \geq \delta\right) = 0 \qquad \forall \delta > 0.$$

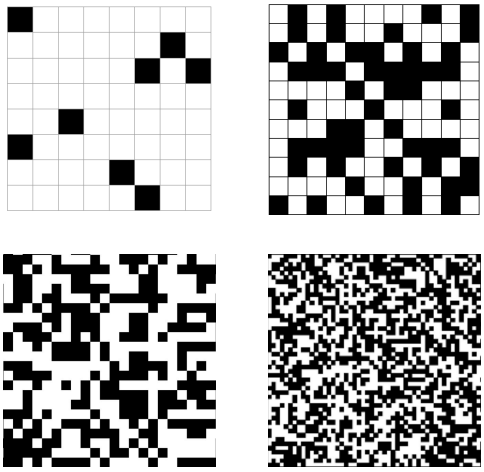


FIGURE 1. Some examples of random checkerboards. The random (closed) subset $\omega_\varepsilon^n \subset [0,1]^2$ is the union of (closed) black squares.

Theorem 2 is proved in Section 3. As mentioned above, this issue has emerged following a question by Nicolas Burq. In [3, 5], the authors also consider checkerboard domains, as above, but not in a random framework. As a consequence of their analysis, given any $T > 0$, any $\varepsilon \in [0,1]$ and any $n \in \mathbb{N}^*$ fixed, if all geodesic rays of length T, either meet the interior of ω_ε^n (i.e.the interior of some black square), or follow for some positive time one of the sides of a black square on the left and for some positive time one of the sides of a black square (possibly the same) on the right, then $C_T(\omega_\varepsilon^n) > 0$, i.e.the wave equation on the torus $M = \mathbb{T}^2$ is observable on ω_ε^n in time T.

Let $T > 0$ and let $\varepsilon \in (0,1]$ be arbitrary. According to Theorem 2, for n large enough, almost every subset ω_ε^n (constructed randomly as above) is such that $\ell^T(\omega_\varepsilon^n) > 0$. This implies that every geodesic ray, that is neither horizontal nor vertical, meets the interior of ω_ε^n within time T, and that every horizontal or vertical geodesic ray meets the closed subset ω_ε^n within time T (for some positive time, not less than $\ell^T(\omega_\varepsilon^n)$). In the latter case, moreover, by construction of the random set ω_ε^n, the probability that vertical grazing rays follow for some positive time one of the sides of a black square on the left and for some positive time one of the sides of a black square on the right, converges to 1 as $n \to +\infty$.

All in all, combining Theorem 2, the result of [3, 5] and the above reasoning, we have the following consequence in terms of observability of the wave equation.

Corollary 2. *Given any $T > 0$ and $\varepsilon \in (0,1]$, may they be arbitrarily small, the probability that the wave equation on the torus $M = \mathbb{T}^2$ be observable on ω_ε^n in time T tends to 1 as $n \to +\infty$.*

In other words, observability in (any) finite time is almost surely true for large n, despite the fact that the measure of ω_ε^n may be very small!

Note that, for $\varepsilon > 1/2$, almost sure observability follows from Corollary 1 (indeed, the random sets constructed above are piecewise C^1 and thus Theorem 1 can be applied). But the result is more striking when $|\omega_\varepsilon^n|$ is small.

Note also Theorem 2 provides an answer to an issue raised in [6], which we formulate in terms of an optimal shape design problem in the next corollary.

Corollary 3. *For every $\varepsilon \in [0,1]$, we have*

$$\sup_{|\omega| \leq \varepsilon} \ell^T(\omega) = \varepsilon,$$

where the supremum is taken over all possible measurable subsets ω of $M = \mathbb{T}^2$ having a Lipschitz boundary.

Corollary 3 is proved in Section 3.4.

We finish this section by a comment on possible generalizations of Theorem 2. Some of the steps of its proof remain valid for any closed Riemannian manifold, like the fact that it suffices to prove the theorem for T small and thus, we expect that, to some extent, the result is purely local. However, in some other steps we instrumentally use the fact that we are dealing with a regular checkerboard in the square. Extending the result to general manifolds, even in dimension two, is an open issue.

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2. Additional comments and proof of Theorem 1

2.1. **Comments on Theorem 1.** Theorem 1 states that, given any $T > 0$ and any measurable subset ω of M, we have

$$\ell^T(\overline{\omega}) \leq \frac{1}{2}\left(\ell^T(\mathring{\omega}) + 1\right)$$

under the two following sufficient assumptions: (i) the metric g is C^2; (ii) ω is an embedded C^1 submanifold of M with boundary if $\dim M \geq 3$ and is piecewise C^1 if $\dim M = 2$.

Remark 1. Assumption (ii) may be weakened as follows:

- If M is of dimension 2, it suffices to assume that ω is piecewise C^1. More precisely, we assume that ω is a C^1 *stratified* submanifold of M (in the sense of Whitney).
- In any dimension, the following much more general assumption is sufficient: given any grazing ray γ, for almost every $t \in [0,T]$ such that $\gamma(t) \in \partial\omega$, the subdifferential at $\gamma(t)$ of $\partial\omega \cap \gamma(\cdot)^\perp$ is a singleton. This is the case under the (much stronger) assumption that ω be geodesically convex.

Comments. It is interesting to note that the assumptions made in Theorem 1 are essentially sharp. Remarks are in order.

- The inequality (4) gives a quantitative measure of the discrepancy that can happen for ℓ^T when we take the closure of a measurable subset ω or, conversely, when we take the interior (this is the sense of Corollary 1). The inequality is sharp, as shown by the example already discussed above: take $M = \mathbb{S}^2$ and ω the open Northern hemisphere; then $\ell^T(\omega) = 0$ for every $T > 0$ and $\ell^{2\pi}(\overline{\omega}) = 1/2$ for $T = 2\pi$. Hence, here, (4) is an equality.
- As a variant, take ω which is the union of the open Northern hemisphere and of a Southern spherical cap, i.e.a portion of the open Southern hemisphere limited by a given latitude $-\varepsilon < 0$. Then we have as well $\ell^T(\omega) = 0$ for every $T > 0$ and $\ell^{2\pi}(\overline{\omega}) = 1/2$ for $T = 2\pi$.
- Note that, taking $\varepsilon = 0$ in the previous example (i.e.ω is the unit sphere $M = \mathbb{S}^2$ minus the equator), we have $\ell^T(\omega) = 0$ and $\ell^T(\overline{\omega}) = 1$ for every $T > 0$ and thus (4) fails. But here, ω is not an embedded C^1 submanifold of M with boundary: Assumption (ii) (which implies local separation between $\mathring{\omega}$ and $M \setminus \overline{\omega}$) is not satisfied. More generally, the result does not apply to any subset ω that is M minus a countable number of rays. This is as well the case when one considers any subset ω that is dense and of empty interior (one has $\ell^T(\mathring{\omega}) = 0$ and $\ell^T(\overline{\omega}) = 1$ for every $T > 0$). This shows that the discrepancy $1/2$ is only valid under some regularity assumptions on ω.
- There is no discrepancy in the absence of geodesic rays grazing ω, i.e.$\ell^T(\overline{\omega}) = \ell^T(\mathring{\omega})$.
- The result fails in general if $\partial\omega$ is piecewise C^1 only, on a manifold M is of dimension $n \geq 3$. Here is a counterexample.

  Let γ be a geodesic ray. If $T > 0$ is small enough, it has no conjugate point. In a local chart, we have $\gamma(t) = (t, 0, \ldots, 0)$ (see the proof of Theorem 1). Now, using this local chart we define a subset ω of M as follows: the section of $\partial\omega$ with the vertical hyperplane $\gamma(\cdot)^\perp$ is locally equal to this entire hyperplane minus a cone of vertex $\gamma(t)$ with small angle $2\pi\varepsilon > 0$, less than $\pi/4$ for instance (see Figure 2).

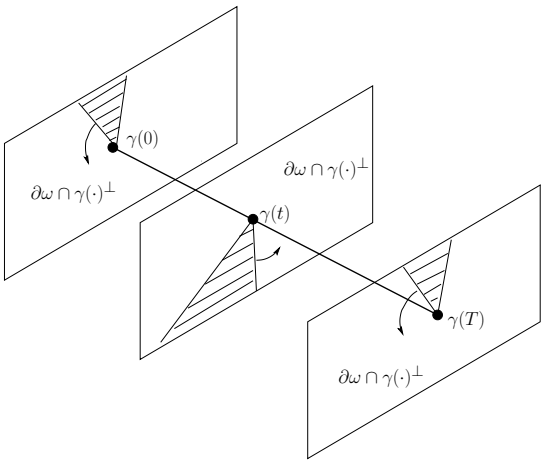


Figure 2. Locally around $\gamma(t)$, $\partial\omega \cap \gamma(t)^\perp$ is the complement of the hatched area.

Now, we assume that, as $t > 0$ increases, these sections rotate with such a speed that, along $[0, T]$, the entire vertical hyperplane is scanned by the section with ω. If the speed of rotation is exactly $T/2\pi$ then it can be proved that $\ell^T(\mathring{\omega}) = 0$ and $\ell^T(\overline{\omega}) = 1 - 2\varepsilon$.

This example shows that Assumption (ii), or its generalization given in Remark 1, cannot be weakened too much. The idea here is to consider a subset ω such that the section of $\partial\omega$ with the vertical hyperplane $\gamma(\cdot)^\perp$ has locally the shape of the hypograph of an absolute value, which is rotating along $\gamma(\cdot)$.

Similar examples can as well be designed with checkerboard-shaped domains ω, thus underlining that in [3, 5] it was important to consider checkerboards in dimension 2.

- Surprisingly, the result is wrong if the metric g is not C^2. A counterexample is the following.

  Let M be a pill-shaped two-dimensional manifold given by the union of a cylinder of finite length, at the extremities of which we glue two hemispheres (domain also obtained by rotating a 2D stadium in $\mathbb{R}^3$ around its longest symmetry axis; or, take the unit sphere in $\mathbb{R}^3$, cut it at the equator, separate the two hemispheres and glue them with, in between, a cylinder of arbitrary length), and endow it with the induced Euclidean metric (see Figure 3).

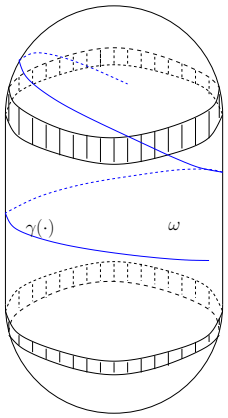


FIGURE 3. M is pill-shaped and ω is the complement of the hatched area.

Then the metric is not C^2 at the gluing circles. Now, take ω defined as the union of the open cylinder with two open spherical caps (i.e.the union of the two hemispheres of which we remove latitudes between 0 and some $\varepsilon > 0$). Then $\ell^T(\omega) = 0$ for every $T > 0$, because ω does not contain the rays consisting of the circles at the extremities of the cylinder. In contrast, $\ell^T(\overline{\omega})$ may be arbitrarily close to 1 as T is large enough and ε is small enough, and thus (4) fails. This is because any ray of M spending a time π in $M \setminus \omega$ spends then much time over the cylinder.

This shows that Assumption (i) is sharp. In the above example, the metric is only $C^{1,1}$.

The example above is rather counter-intuitive. The assumption of a C^2 metric implies in some sense a global result on geodesic rays.

Our proof, given in Section 2.2 hereafter, uses only elementary arguments of Riemannian geometry. It essentially relies on Lemma 2, in which we establish that, given a grazing ray (i.e.a ray propagating in $\partial\omega$), thanks to our assumption on ω, we can always construct neighbor rays, one of which being inside ω and the other being outside of ω for all times.

2.2. **Proof of Theorem 1.** Without loss of generality, we take $\omega \subset M$ open. We will use several well known facts of Riemannian geometry, for which we refer to [2].

Lemma 1. *There exists $\gamma \in \Gamma$ such that $\ell^T(\omega) = \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\,dt$, i.e.the infimum in the definition* (1) *of $\ell^T(\omega)$ is reached.*

Proof. Let $(\gamma_k)_{k\in\mathbb{N}}$ be a sequence of geodesic rays such that $\frac{1}{T}\int_0^T \chi_\omega(\gamma_k(t))\,dt \to \ell^T(\omega)$. By compactness of geodesics, taking a subsequence if necessary, $\gamma_k(\cdot)$ converges uniformly to some geodesic ray $\gamma(\cdot)$ on $[0, T]$.

Let $t \in [0, T]$ be arbitrary. If $\gamma(t) \in \omega$ then for k large enough we have $\gamma_k(t) \in \omega$, and thus $1 = \chi_\omega(\gamma(t)) \leq \chi_\omega(\gamma_k(t)) = 1$. If $\gamma(t) \in M \setminus \omega$ then $0 = \chi_\omega(\gamma(t)) \leq \chi_\omega(\gamma_k(t))$ for any k. In all cases, we have obtained the inequality

$$\chi_\omega(\gamma(t)) \leq \liminf_{k\to+\infty} \chi_\omega(\gamma_k(t))$$

for every $t \in [0,T]$. By the Fatou lemma, we infer that

$$\ell^T(\omega) \le \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\,dt \le \frac{1}{T}\int_0^T \liminf_{k\to+\infty} \chi_\omega(\gamma_k(t))\,dt \le \liminf_{k\to+\infty} \frac{1}{T}\int_0^T \chi_\omega(\gamma_k(t))\,dt = \ell^T(\omega).$$

The lemma follows. □

If the ray γ given by Lemma 1 is not grazing ω, i.e.if $\int_0^T \chi_{\partial\omega}(\gamma(t))\,dt = 0$, then $\int_0^T \chi_\omega(\gamma(t))\,dt = \int_0^T \chi_{\overline{\omega}}(\gamma(t))\,dt$ and thus $\ell^T(\overline{\omega}) \le \frac{1}{T}\int_0^T \chi_{\overline{\omega}}(\gamma(t))\,dt \le \ell^T(\omega)$ and hence $\ell^T(\overline{\omega}) = \ell^T(\omega)$. So in this case there is nothing to prove.

In what follows we assume that the ray γ given by Lemma 1 is grazing, i.e.$\int_0^T \chi_{\partial\omega}(\gamma(t))\,dt > 0$. Assume that $\gamma(t) = \pi\circ\varphi_t(x_0,\xi_0)$ with $x_0 \in M$ and $\xi_0 \in S^*_{x_0}M$. Here, $S^*_{x_0}M$ denotes the unit cotangent bundle at x_0 (i.e.$\|\xi_0\|_{g^\star} = 1$, where g^* is the cometric), $(\varphi_t)_{t\in\mathbb{R}}$ is the geodesic flow on S^*M and $\pi : S^*M \to M$ is the canonical projection.

Lemma 2. *There exists a continuous path of points $s \mapsto x_s \in M$, passing through x_0 at $s = 0$, such that, setting $\gamma_s(t) = \pi\circ\varphi_t(x_s,\xi_0)$, we have*

$$\lim_{s\to 0} \left(\chi_{\overline{\omega}}(\gamma_s(t)) + \chi_{\overline{\omega}}(\gamma_{-s}(t))\right) = 1 \tag{5}$$

for almost every $t \in [0,T]$ such that $\gamma(t) \in \partial\omega$.

Proof. To prove this fact, we assume that, in a local chart, $\gamma(t) = (t,0,\dots,0)$. This is true at least in a neighborhood of $x_0 = \gamma(0) = 0$, and this holds true along $\gamma(\cdot)$ as long as there is no conjugate point. We also assume that, in this chart, any other geodesic ray starting at $(0,x_2^0,\dots,x_n^0)$ in a neighborhood of $\gamma(0) = (0,\dots,0)$, with codirection ξ_0, is given by $(t,x_2^0,\dots,x_d^0)$ (projection onto M of the extremal field). Here, we have set $d = \dim M$. This classical construction of the so-called *extremal field* can actually be done on any subinterval of $[0,T]$ along which there is no conjugate point. Note that the set of conjugate times along $[0,T]$ is of Lebesgue measure zero.[1] Let us search an appropriate $(d-1)$-tuple $(x_2^0,\dots,x_d^0) \in \mathbb{R}^{d-1}\setminus\{0\}$ such that the family of points $x_s = (0,sx_2^0,\dots,sx_d^0)$, $s \in (-1,1)$, gives (5). Note that the geodesic ray starting at (x_s,ξ_0) is $\gamma_s(t) = (t,sx_2^0,\dots,sx_d^0)$ in the local chart.

In what follows, we set $N = \partial\omega = \overline{\omega}\setminus\mathring{\omega} = \overline{\omega}\setminus\omega$ (ω is open). By assumption, ω is an embedded C^1 submanifold of M with boundary and one has $\dim N = d-1$. By assumption, in a neighborhood U of any point of N, the set $N\cap U$ is a codimension-one hypersurface of M, written as $F = 0$ with $F : U \to \mathbb{R}$ of class C^1, which is separating ω and $M\setminus\omega$ in the sense that $\omega\cap U = \{F < 0\}$, $N = \{F = 0\}$ and $M\setminus\omega = \{F \ge 0\}$.

It suffices to prove that, for almost every time t at which $\gamma_0(t) = \gamma(t) \in N$ and $\dot\gamma(t) \in T_{\gamma(t)}N$, the points $\gamma_s(t)$ and $\gamma_{-s}(t)$ are on different sides with respect to the (locally) separating manifold N for s small enough.

This is obvious when γ is transverse to N. We set

$$\Omega = \{t \in [0,T] \mid \gamma(t) \in N,\ \dot\gamma(t) \in T_{\gamma(t)}N\}.$$

It is a closed subset of $[0,T]$. Let $t \in \Omega$. In the local chart the tangent space $T_{\gamma(t)}N$ is an hyperplane of $\mathbb{R}^n$ containing the line $\mathbb{R}(1,0,\dots,0)$. Its projection onto $\{0\}\times\mathbb{R}^{d-1}$ (the hyperplane orthogonal to the line $\gamma(\cdot)$) is an hyperplane of $\{0\}\times\mathbb{R}^{d-1}$, of normal vector $(0,v(t))$ with $v(t) \in \mathbb{R}^{d-1}$ of Euclidean norm 1. Since only the direction of $v(t)$ is important, we assume that $v(t) \in \mathbb{P}^{d-2}(\mathbb{R})$, the projective space.

We claim that:

There exists $V \in \mathbb{P}^{d-2}(\mathbb{R})$ such that $\langle V, v(t)\rangle \neq 0$ for almost every $t \in \Omega$.

With this result, setting $V = (x_2^0,\dots,x_d^0)$, the points x_s defined above give the lemma.

Let us now prove the claim. We define $A = \{(t,V) \in \Omega\times\mathbb{P}^{d-2}(\mathbb{R}) \mid \langle V, v(t)\rangle = 0\}$. By definition, given $(t,V) \in \Omega\times\mathbb{P}^{d-2}(\mathbb{R})$ we have $\chi_A(t,V) = 1$ when $V \in v(t)^\perp$. Since $v(t)^\perp \cap \mathbb{P}^{d-2}(\mathbb{R})$ is of codimension one in $\mathbb{P}^{d-2}(\mathbb{R})$, we have $\int_{\mathbb{P}^{d-2}(\mathbb{R})} \chi_A(t,V)\,d\mathcal{H}^{d-2} = 0$ for every $t \in \Omega$, where we have endowed $\mathbb{P}^{d-2}(\mathbb{R})$ with the Hausdorff measure $\mathcal{H}^{n-2}$. Therefore, by the Fubini theorem,

$$0 = \int_\Omega \int_{\mathbb{P}^{d-2}(\mathbb{R})} \chi_A(t,V)\,d\mathcal{H}^{d-2}\,dt = \int_{\mathbb{P}^{d-2}(\mathbb{R})} \int_\Omega \chi_A(t,V)\,dt\,d\mathcal{H}^{d-2}$$

and thus $\int_\Omega \chi_A(t,V)\,dt = 0$ for almost every $V \in \mathbb{P}^{d-2}(\mathbb{R})$. Fixing such a V, it follows that $\chi_A(t,V) = 0$ for almost every $t \in \Omega$, and the claim is proved. □

[1] This is a general fact in Riemannian geometry. Indeed, a conjugate time is a time at which a non-zero Jacobi field vanishes. Since Jacobi fields are solutions of a second-order ordinary differential equation, such times must be isolated, for otherwise the Jacobi field would vanish at the second order and thus would be identically zero.

In view of proving Remark 1, note that the argument above still works in dimension 2 with ω piecewise C^1 (but not in dimension greater than or equal to 3: see the counterexample given in Section 1). In more general, in any dimension, the argument above still works if ω is such that, for almost every time t, the subdifferential at $\gamma(t)$ of $\partial\omega \cap \gamma(\cdot)^\perp$ is a singleton.

At this step, we have embedded the ray γ given by Lemma 1 into a family of rays γ_s which enjoy a kind of transversality property with respect to $N = \partial\omega$. Let us consider the partition

$$[0,T] = A_1 \cup A_2 \cup A_3$$

into three disjoint measurable sets, with

$$\begin{aligned} A_1 &= \{t \in [0,T] \mid \gamma(t) \in \omega\},\\ A_2 &= \{t \in [0,T] \mid \gamma(t) \in M \setminus \overline{\omega}\},\\ A_3 &= \{t \in [0,T] \mid \gamma(t) \in \partial\omega\}. \end{aligned}$$

Since $\gamma_s(\cdot)$ converges uniformly to $\gamma(\cdot)$ as $s \to 0$ and since ω and $M \setminus \overline{\omega}$ are open, we have:

$$\begin{aligned} &\lim_{s\to 0} \left(\chi_{\overline{\omega}}(\gamma_s(t)) + \chi_{\overline{\omega}}(\gamma_{-s}(t))\right) = 2 \quad \text{for every } t \in A_1;\\ &\lim_{s\to 0} \left(\chi_{\overline{\omega}}(\gamma_s(t)) + \chi_{\overline{\omega}}(\gamma_{-s}(t))\right) = 0 \quad \text{for every } t \in A_2;\\ &\lim_{s\to 0} \left(\chi_{\overline{\omega}}(\gamma_s(t)) + \chi_{\overline{\omega}}(\gamma_{-s}(t))\right) = 1 \quad \text{for almost every } t \in A_3 \text{ (by Lemma 2).} \end{aligned}$$

By the Lebesgue dominated convergence theorem, we infer that

$$\lim_{s\to 0} \int_0^T \left(\chi_{\overline{\omega}}(\gamma_s(t)) + \chi_{\overline{\omega}}(\gamma_{-s}(t))\right) dt = 2|A_1| + |A_3|.$$

Now, on the one part, by the first step we have $\frac{1}{T}|A_1| = \frac{1}{T}\int_0^T \chi_\omega(\gamma(t)) = \ell^T(\omega)$. On the other part, since A_1 and A_3 are disjoint we have $\frac{1}{T}(|A_1| + |A_3|) \leq 1$. Hence

$$\lim_{s\to 0} \frac{1}{T} \int_0^T \left(\chi_{\overline{\omega}}(\gamma_s(t)) + \chi_{\overline{\omega}}(\gamma_{-s}(t))\right) dt \leq \ell^T(\omega) + 1.$$

Since $\ell^T(\overline{\omega}) \leq \frac{1}{T}\int_0^T \chi_{\overline{\omega}}(\gamma_{\pm s}(t))dt$ for every s by definition, we infer that $2\ell^T(\overline{\omega}) \leq \ell^T(\omega) + 1$. Theorem 1 is proved.

3. Proof of Theorem 2

Theorem 2 states that, given any $T > 0$ and any $\varepsilon \in [0,1]$, the random variable $\ell^T(\omega_\varepsilon^n)$ converges in probability to ε as $n \to +\infty$, i.e.

$$\lim_{n\to+\infty} \mathbb{P}\left(|\ell^T(\omega_\varepsilon^n) - \varepsilon| \geq \delta\right) = 0 \qquad \forall \delta > 0.$$

This section is organized as follows. We make a preliminary remark in Section 3.1. In Section 3.2, we give the successive steps of the proof, involving intermediate lemmas that are proved. One of the main ingredients of the proof of Theorem 2 is a large deviation property which is established in Section 3.3. In Section 3.4, we also provide a proof of Corollary 3.

3.1. Preliminaries.

For any geodesic ray $\gamma \in \Gamma$ and any measurable subset $\omega \subset M$, we set

$$m_\gamma^T(\omega) = \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\, dt,$$

so that

$$\ell^T(\omega) = \inf_{\gamma\in\Gamma} m_\gamma^T(\omega). \tag{6}$$

Lemma 3. *Given any measurable closed subset $\omega \subset M$, the mapping $\Gamma \ni \gamma \mapsto m_\gamma^T(\omega)$ is upper semi-continuous.*

Given any ω that is a union of closed squares from the grid $\mathcal{G}^n$ with n fixed, the mapping $\Gamma \ni \gamma \mapsto m_\gamma^T(\omega)$ is continuous at every $\gamma \in \Gamma$ that is neither horizontal nor vertical, or, that is horizontal or vertical but meets no corner (by definition, a corner is a point $(i/n, j/n)$ in $[0,1]^2$, for some $(i,j) \in \{0,\dots,n\}^2$).

Proof. Let $\gamma \in \Gamma$ and let $(\gamma_k)_{k\in\mathbb{N}}$ be a sequence of Γ converging to γ. Let us prove that

$$m_\gamma^T(\omega) \geq \limsup_{k\to+\infty} m_{\gamma_k}^T(\omega). \tag{7}$$

Let $t \in [0,T]$. If $\gamma(t) \in \omega$ then $\chi_\omega(\gamma(t)) \geq \chi_\omega(\gamma_k(t))$ for every k. If $\gamma(t) \in M \setminus \omega$ then, since $M \setminus \omega$ is open, $\gamma_k(t) \in M \setminus \omega$ for k large enough and thus $\chi_\omega(\gamma(t)) = \chi_\omega(\gamma_k(t)) = 0$. Now (7) follows.

Let us prove the second part of the lemma. Let $\gamma \in \Gamma$ that is neither horizontal nor vertical, or, that is horizontal or vertical but meets no corner. Let $(\gamma_k)_{k\in\mathbb{N}}$ be a sequence of geodesic rays converging to $\gamma \in \Gamma$. Writing

$$\chi_\omega = \sum_{i,j=1}^{n} \tau(i,j)\chi_{c_{ij}^n},$$

where $\tau(i,j) = 1$ if $c_{ij}^n \subset \omega$ and $\tau(i,j) = 0$ otherwise, and noting that

$$m_{\gamma_k}^T(\omega) = \sum_{i,j=1}^{n} \tau(i,j) m_{\gamma_k}^T(c_{ij}^n),$$

we have to prove that $m_{\gamma_k}^T(c_{ij}^n)$ converges to $m_\gamma^T(c_{ij}^n)$ as $k \to +\infty$. This follows from the dominated convergence theorem and from the fact that $(\chi_{c_{ij}^n}(\gamma_k))_{k\in\mathbb{N}}$ converges almost everywhere to $\chi_{c_{ij}^n}(\gamma)$. The latter claim can be shown by distinguishing between two cases: if $\gamma(t) \in \mathring{c}_{ij}^n$ then for k large enough we have $\gamma_k(t) \in \mathring{c}_{ij}^n$ and $\chi_{c_{ij}^n}(\gamma_k(t)) = \chi_{c_{ij}^n}(\gamma(t)) = 1$. The same conclusion remains true if $\gamma(t) \notin c_{ij}^n$. Since the set of t such that $\gamma(t) \in \partial c_{ij}^n$ is finite (this follows from the assumptions on γ), the lemma follows. □

Note that $\gamma \mapsto m_\gamma^T$ may fail to be continuous at some $\gamma \in \Gamma$ that is horizontal or vertical and meets a corner. For such geodesic rays γ, we will see that it is relevant to define the following quantity. Given any $\gamma \in \Gamma$, we set $\widetilde{m}_\gamma^T(\omega) = m_\gamma^T(\omega)$ if γ is neither horizontal nor vertical, or is horizontal or vertical but meets no corner. When γ is horizontal or vertical and meets a corner, we set

$$\widetilde{m}_\gamma^T(\omega) = \inf_{(\gamma_k')_{k\in\mathbb{N}}} \liminf_{k\to+\infty} m_{\gamma_k'}^T(\omega), \tag{8}$$

where the infimum is taken over the set of sequences of geodesic rays $(\gamma_k')_{k\in\mathbb{N}}$ converging to γ such that, for every $k \in \mathbb{N}$, $\gamma_k'([0,T])$ is obtained by rotating $\gamma([0,T])$ around a corner of the grid through which γ passes. Of course, we have $\ell^T(\omega) \le \widetilde{m}_\gamma^T(\omega) \le m_\gamma^T(\omega)$ and thus

$$\ell^T(\omega) = \inf_{\gamma\in\Gamma} \widetilde{m}_\gamma^T(\omega).$$

Note that $\gamma \mapsto \widetilde{m}_\gamma^T(\omega)$ may also fail to be continuous. We will see that this quantity $\widetilde{m}_\gamma^T(\omega)$ is important to obtain the inequality (25) in the proof of Lemma 5, item (b), and in order to treat the case of horizontal or vertical rays in the proof of Lemma 6.

3.2. **Proof of Theorem 2.** For every $(i,j) \in \{1,\dots,n\}^2$, let X_{ij}^n be the random variable equal to 1 whenever $c_{ij}^n \subset \omega_\varepsilon^n$ and 0 otherwise. Recall that, assuming that all square cells of the grid are initially white, when ranging over the grid, for each cell we randomly darken the cell with a probability ε (Bernoulli law) and in this case we set $X_{ij}^n = 1$; otherwise we let $X_{ij}^n = 0$. By construction, the random laws $(X_{ij}^n)_{1\le i,j\le n}$ are independent and identically distributed (i.i.d.), of expectation ε and of variance $\varepsilon(1-\varepsilon)$.

Given any fixed geodesic ray $\gamma \in \Gamma$, we denote by $t_{ij}^n(\gamma)$ the time spent by γ in the square cell c_{ij}^n. We have $\sum_{i,j=1}^n t_{ij}^n(\gamma) = T$. It may happen that γ crosses several times (at most $T+1$ times) the square cell c_{ij}^n, and in this case $t_{ij}^n(\gamma)$ is the sum of several passage times; since the maximal time spent in c_{ij}^n by a ray in one passage is $\sqrt{2}/n$, we have $t_{ij}^n(\gamma) \le (T+1)\sqrt{2}/n$. Summing up, we have

$$\forall (i,j) \in \{1,\dots,n\}^2 \qquad \frac{t_{ij}^n(\gamma)}{T} \le \frac{T+1}{T}\,\frac{\sqrt{2}}{n} \qquad \text{and} \qquad \sum_{i,j=1}^{n} \frac{t_{ij}^n(\gamma)}{T} = 1 \tag{9}$$

and we also note that

$$m_\gamma^T(\omega_\varepsilon^n) = \frac{1}{T}\int_0^T \chi_{\omega_\varepsilon^n}(\gamma(t))\,dt = \sum_{i,j=1}^{n} \frac{t_{ij}^n(\gamma)}{T}\,X_{ij}^n. \tag{10}$$

In particular, the random variable $m_\gamma^T(\omega_\varepsilon^n)$ is a weighted sum of independent Bernoulli laws, and thus its expectation is

$$\mathbb{E}\, m_\gamma^T(\omega_\varepsilon^n) = \varepsilon \tag{11}$$

and its variance is

$$\operatorname{Var} m_\gamma^T(\omega_\varepsilon^n) = \sum_{i,j=1}^{n} \Big(\frac{t_{ij}^n(\gamma)}{T}\Big)^2 \varepsilon(1-\varepsilon) \le \frac{T+1}{T}\,\frac{\sqrt{2}}{n}\,\varepsilon(1-\varepsilon), \tag{12}$$

where we have used (9) and the fact that $0 \le {}^{t_{ij}^n(\gamma)}/_T \le 1$.

To prove Theorem 2, we have to prove that, given any $T > 0$ and any $\varepsilon \in [0,1]$, for every $\delta > 0$ we have

(i) $\lim_{n\to+\infty} \mathbb{P}\left(\ell^T(\omega_\varepsilon^n) \le \varepsilon + \delta\right) = 1$,

(ii) $\lim_{n\to+\infty} \mathbb{P}\left(\ell^T(\omega_\varepsilon^n) \geq \varepsilon - \delta\right) = 1$, or equivalently, $\lim_{n\to+\infty} \mathbb{P}\left(\ell^T(\omega_\varepsilon^n) \leq \varepsilon - \delta\right) = 0$.

Proof of (i). Using (6) and (10), we have

$$\ell^T(\omega_\varepsilon^n) \leq m_\gamma^T(\omega_\varepsilon^n). \tag{13}$$

Applying the Bienaymé-Tchebychev inequality to the random variable $m_\gamma^T(\omega_\varepsilon^n)$, we have

$$\mathbb{P}\left(|m_\gamma^T(\omega_\varepsilon^n) - \varepsilon| \geq \delta\right) \leq \frac{T+1}{T}\,\frac{\sqrt{2}}{n}\,\frac{\varepsilon(1-\varepsilon)}{\delta^2} \qquad \forall \gamma \in \Gamma$$

and thus, using (11) and (12),

$$\lim_{n\to+\infty} \mathbb{P}\left(m_\gamma^T(\omega_\varepsilon^n) \leq \varepsilon + \delta\right) = 1. \tag{14}$$

Finally, (i) follows from (13) and (14).

Proof of (ii). Establishing (ii) is much more difficult. We proceed in several steps, by proving the following successive lemmas that are in order.

Lemma 4. *If* (ii) *is true for every* $T \in (0,1)$*, then it is true for every* $T > 0$.

Thanks to Lemma 4 (proved below), we now assume that $0 < T < 1$. This has the following pleasant consequence: any geodesic ray $\gamma \in \Gamma$ crosses a given cell c_{ij} at most one time over $[0,T]$, i.e.$\{t \in [0,T] \mid \gamma(t) \in c_{ij}\}$ is connected. This will make easier the computation of crossing times, in the proofs of the forthcoming lemmas.

Let us introduce some notations. Let Γ^0, Γ^1 and Γ^2 be the sets of geodesic rays of $M = \mathbb{T}^2$ meeting respectively zero, at least one and at least two corners of the grid $\mathcal{G}^n$ (by definition, a corner is a point $(i/n, j/n)$ in $[0,1]^2$, for some $(i,j) \in \{0,\dots,n\}^2$). Note that $\Gamma^2 \subset \Gamma^1$.

Given any $\omega \subset M$ that is the union of disjoint closed square cells of $\mathcal{G}^n$, we have

$$\ell^T(\omega) = \inf_{\gamma\in\Gamma^0\cup\Gamma^1} m_\gamma^T(\omega).$$

We also define

$$\ell_{\Gamma^1}^T(\omega) = \inf_{\gamma\in\Gamma^1} m_\gamma^T(\omega), \qquad \ell_{\Gamma^2}^T(\omega) = \inf_{\gamma\in\Gamma^2} m_\gamma^T(\omega).$$

Of course, we have

$$\ell^T(\omega) \leq \ell_{\Gamma^1}^T(\omega) \leq \ell_{\Gamma^2}^T(\omega).$$

Lemma 5. *There exists* $C > 0$ *such that, for every* $n \in \mathbb{N}^*$*, for every subset* ω *of* $[0,1]^2$ *that is a union of square cells of the grid* $\mathcal{G}_n$*, we have:*

(a) $\ell_{\Gamma^1}^T(\omega) \leq \ell^T(\omega) + C/n$;
(b) $\ell_{\Gamma^2}^T(\omega) \leq \ell_{\Gamma^1}^T(\omega) + C/n$.

Lemma 6. *We have* $\lim_{n\to+\infty} \mathbb{P}\left(\ell_{\Gamma^2}^T(\omega_\varepsilon^n) \leq \varepsilon - \delta\right) = 0$.

Finally, (ii) follows from the above lemmas, that are proved hereafter.

3.2.1. *Proof of Lemma 4.* Let $T \geq 1$ and let $m \in \mathbb{N}^*$ be such that $T' = T/m \in (0,1)$. Let us prove that

$$\ell^T(\omega_\varepsilon^n) \geq \ell^{T'}(\omega_\varepsilon^n). \tag{15}$$

Given any $\rho > 0$, let γ be a geodesic ray such that

$$\ell^T(\omega_\varepsilon^n) + \rho \geq \frac{1}{T}\int_0^T \chi_{\omega_\varepsilon^n}(\gamma(t))\,dt.$$

Setting $\gamma_k(\cdot) = \gamma(kT' + \cdot)$ for every $k \in \{0,\dots,m-1\}$, we have

$$\ell^T(\omega_\varepsilon^n) + \rho \geq \frac{1}{T}\int_0^T \chi_{\omega_\varepsilon^n}(\gamma(t))\,dt = \frac{1}{T}\sum_{k=0}^{m-1}\int_0^{T'} \chi_{\omega_\varepsilon^n}(\gamma_k(t))\,dt$$
$$\geq \frac{1}{T}\sum_{k=0}^{m-1} T'\ell^{T'}(\omega_\varepsilon^n) = \ell^{T'}(\omega_\varepsilon^n).$$

Letting ρ tend to 0, we obtain (15).

Since $0 < T' < 1$, (ii) is true for this final time T', i.e.

$$\lim_{n\to+\infty} \mathbb{P}\left(\ell^{T'}(\omega_\varepsilon^n) \geq \varepsilon - \delta\right) = 1.$$

Therefore, using (15), we obtain (ii) for the final time T.

3.2.2. *Proof of Lemma 5.*

Proof of item (a). Let $\gamma \in \Gamma$ be such that

$$\ell^T(\omega) \le m_\gamma^T(\omega) < \ell^T(\omega) + \frac{1}{nT}. \tag{16}$$

If $\gamma \in \Gamma^1$ then we are done. Hence, in what follows we assume that $\gamma \in \Gamma^0$, i.e.that γ meets no corner of the grid.

Without loss of generality, we can assume that the ray γ is neither horizontal nor vertical. Indeed, if γ is horizontal or vertical, since γ meets no corner, it follows from Lemma 3 that m_γ^T is continuous at γ. Hence, it is possible to rotate slightly γ so that γ is neither horizontal nor vertical and still satisfies (16).

Let $\boldsymbol{n} \in \mathbb{R}^2$ be a unit vector orthogonal to $\gamma'(0)$. For every $s \in \mathbb{R}$, we denote by $\mathcal{T}_{s\boldsymbol{n}}$ the translation of vector $s\boldsymbol{n}$ and we define the translated geodesic ray $\gamma_s = \mathcal{T}_{s\boldsymbol{n}} \circ \gamma$ (which is neither horizontal nor vertical). By continuity, γ_s meets the same square cells as γ if $|s|$ is small enough. Let $I(\gamma)$ denote the subset of all pairs $(i,j) \in \{1,\dots,n\}^2$ such that γ crosses the cell squares c_{ij}^n. For $|s|$ small enough, we have

$$m_{\gamma_s}^T(\omega) = \sum_{(i,j)\in I(\gamma)} \frac{t_{ij}^n(\gamma_s)}{T}\, X_{ij}^n,$$

where $t_{ij}^n(\gamma_s)$ is the time spent by γ_s in c_{ij}^n. Denoting by $I'(\gamma)$ the set of $(i,j) \in I(\gamma)$ such that $\gamma(0), \gamma(T) \notin \mathring{c}_{ij}^n$ (note that $\#I(\gamma) - 2 \le \#I'(\gamma) \le \#I(\gamma)$ and that these conditions do not depend on s for $|s|$ small enough, hence $I'(\gamma_s) = I'(\gamma)$), an easy geometric argument shows that, for $(i,j) \in I'(\gamma)$, the function $s \mapsto t_{ij}^n(\gamma_s)$ is affine and nonconstant with respect to s. Hence,

$$M_{\gamma_s}(\omega) = \sum_{(i,j)\in I'(\gamma)} \frac{t_{ij}^n(\gamma_s)}{T}\, X_{ij}^n \tag{17}$$

is also an affine function of s. Replacing s by $-s$ if necessary, we infer the existence of a threshold $s_0 > 0$ such that $\gamma_s \in \Gamma^0$ for $s \in [0, s_0)$ and $\gamma_{s_0} \in \Gamma^1$, and the mapping $s \mapsto M_{\gamma_s}(\omega)$ is continuous and nonincreasing on $[0, s_0]$ (continuity is because, for every s, γ_s is neither vertical nor horizontal since it is a translation of γ). Since $\gamma_{s_0} \in \Gamma^1$, we have

$$\ell_{\Gamma^1}^T(\omega) \le m_{\gamma_{s_0}}^T(\omega). \tag{18}$$

Besides, we have, for every $s \in [0, s_0]$,

$$M_{\gamma_s}(\omega) \le m_{\gamma_s}^T(\omega) \le M_{\gamma_s}(\omega) + \sum_{(i,j)\in I(\gamma)\setminus I'(\gamma)} \frac{t_{ij}^n(\gamma_s)}{T}\, X_{ij}^n \le M_{\gamma_s}(\omega) + \frac{2\sqrt{2}}{nT}. \tag{19}$$

Hence, in particular, using that (by Lemma 3) $s \mapsto m_{\gamma_s}^T(\omega)$ is continuous because γ_s is neither horizontal nor vertical,

$$m_{\gamma_{s_0}}^T(\omega) \le M_{\gamma_{s_0}}(\omega) + \frac{2\sqrt{2}}{nT}. \tag{20}$$

Since the mapping $s \mapsto M_{\gamma_s}(\omega)$ is nonincreasing on $[0, s_0]$ and $\gamma_0 = \gamma$, we have $M_{\gamma_{s_0}}(\omega) \le M_\gamma(\omega)$ and thus, using (19),

$$M_{\gamma_{s_0}}(\omega) \le m_\gamma^T(\omega). \tag{21}$$

We finally infer from (16), (18), (20) and (21) that

$$\ell_{\Gamma^1}^T(\omega) < \ell^T(\omega) + \frac{2\sqrt{2}+1}{nT}.$$

The conclusion follows.

Proof of item (b). Let $\gamma \in \Gamma^1$ be such that

$$\ell_{\Gamma^1}^T(\omega) \le m_\gamma^T(\omega) < \ell_{\Gamma^1}^T(\omega) + \frac{1}{n}. \tag{22}$$

If $\gamma \in \Gamma^2$ then we are done. Hence, in what follows we assume that $\gamma \in \Gamma^1 \setminus \Gamma^2$. It suffices to find $\gamma' \in \Gamma^2$ such that

$$m_{\gamma'}^T(\omega) \le m_\gamma^T(\omega) + \frac{C}{n} \tag{23}$$

for some $C > 0$ not depending on γ and n. Indeed, then, it follows from (22) and (23) that

$$\ell_{\Gamma^2}^T(\omega) \le m_{\gamma'}^T(\omega) \le m_\gamma^T(\omega) + \frac{C}{n} < \ell_{\Gamma^1}^T(\omega) + \frac{C+1}{n}$$

and the item is proved.

Let us establish (23). By definition, the geodesic ray γ meets exactly one corner of the grid, denoted by O. Let R_θ be the rotation centered at O with angle θ and let $\gamma_\theta = r_\theta \circ \gamma \in \Gamma$. Recall that $I'(\gamma)$ stands for the set of all pairs $(i,j) \in \{1,\dots,n\}^2$ such that γ_θ crosses c_{ij}^n and $\gamma(0), \gamma(T) \notin \mathring{c}_{ij}^n$: these conditions do not depend on $\theta \in J = (\theta_-, \theta_+)$, where $\theta_- < 0$ and $\theta_+ > 0$ are thresholds such that $\gamma_\theta \notin \Gamma^2$ for every $\theta \in J$ and $\gamma_{\theta_\pm} \in \Gamma^2$. We define the continuous function f on J by $f(\theta) = M_{\gamma_\theta}(\omega)$, defined as in (17) but where γ_θ is obtained by rotation of γ and not by translation. We extend f by continuity on $\bar{J}$. The inequality (19) is valid as well for every $\theta \in \bar{J}$.

Note, however, that the functions $\theta \mapsto m_{\gamma_\theta}^T(\omega)$ and $\theta \mapsto M_{\gamma_\theta}(\omega)$ may fail to be continuous at $\theta_\pm$ (i.e.we may have $f(\theta_\pm) \neq M_{\gamma_\theta}(\omega_\pm)$) whenever $\gamma_{\theta_\pm}$ is horizontal or vertical and meets a corner.

We claim that

$$\min(f(\theta_-), f(\theta_+)) \leq \min_{\theta \in \bar{J}} f(\theta) + \frac{K}{n} \tag{24}$$

for some $K > 0$ not depending on n nor on γ. Admitting temporarily (24), without loss of generality we assume that $f(\theta_-) \leq f(\theta_+)$. Let us prove that

$$\widetilde{m}_{\gamma_{\theta_-}}^T(\omega) \leq f(\theta_-) + \frac{2\sqrt{2}}{nT}, \tag{25}$$

where $\widetilde{m}_\gamma^T(\omega)$ is defined by (8) in Section 3.1. When γ_{θ_-} is neither horizontal nor vertical, or is horizontal or vertical but meets no corner, we have $\widetilde{m}_{\gamma_{\theta_-}}^T(\omega) = m_{\gamma_{\theta_-}}^T(\omega)$ and (25) follows from (19) and from the fact that $f(\theta_-) = M_{\gamma_\theta}(\omega_-)$ (actually, (25) is an equality in this case). When γ_{θ_-} is horizontal or vertical and meets a corner, it is the limit of $\gamma'_k = \gamma_{\theta_- + \frac{1}{k}}$ and then, by definition, $\widetilde{m}_{\gamma_{\theta_-}}^T(\omega) \leq m_{\gamma'_k}^T(\omega)$; but, using (19), we have

$$m_{\gamma'_k}^T(\omega) \leq M_{\gamma'_k}(\omega) + \frac{2\sqrt{2}}{nT} = f(\theta_- + 1/k) + \frac{2\sqrt{2}}{nT},$$

whence (25) by continuity of f.

We infer from (24) and (25) that

$$\widetilde{m}_{\gamma_{\theta_-}}^T(\omega) \leq \min_{\theta \in \bar{J}} f(\theta) + \frac{K}{n} + \frac{2\sqrt{2}}{nT}.$$

Since $\min_{\theta \in \bar{J}} f(\theta) \leq f(0) = M_\gamma(\omega) \leq m_\gamma^T(\omega)$, we obtain $\widetilde{m}_{\gamma_{\theta_-}}^T(\omega) \leq m_\gamma^T(\omega) + K'/n$ with $K' = K + 2\sqrt{2}/T$. By definition of $\widetilde{m}_{\gamma_{\theta_-}}^T(\omega)$ in (8), there exists a geodesic ray γ' such that $\widetilde{m}_{\gamma_{\theta_-}}^T(\omega) + 1/n \geq m_{\gamma'}^T(\omega)$. Therefore, (23) is proved by setting $C = K' + 1$.

It remains to prove (24). Let us first compute f by deriving an explicit expression of $t_{ij}^n(\gamma_\theta)$. We claim that, for $(i,j) \in I(\gamma)$, there exists $(a_{ij}, b_{ij}) \in \{-n,\dots,n\}^2$ such that

$$t_{ij}^n(\gamma_\theta) = \frac{a_{ij}}{n\sin(\theta_0 + \theta)} + \frac{b_{ij}}{n\cos(\theta_0 + \theta)}, \tag{26}$$

where θ_0 is the angle between the horizontal axis and the geodesic ray γ. Without loss of generality we assume that $\theta_0 \in [0, \pi/2]$. Note that $\theta_0 \in (0, \pi/2)$, because otherwise the two conditions $\gamma \in \Gamma_1$ and $\theta_0 \in \{0, \pi/2\}$ (i.e.γ is horizontal or vertical) would imply that $\gamma \in \Gamma_2$, which is not true. This also implies that $J \subset (0, \pi/2)$.

Using the notations of Figure 4 where the corner O is assimilated to the origin, and focusing on the square c_{ij}^n of vertices

$$(\alpha/n, \beta/n),\ (\alpha/n + 1/n, \beta/n),\ (\alpha/n + 1/n, \beta/n + 1/n) \text{ and } (\alpha/n, \beta/n + 1/n),$$

an elementary geometric reasoning gives

$$t_{ij}^n(\gamma_\theta) = \begin{cases} \dfrac{\alpha+1}{n\cos(\theta_0+\theta)} - \dfrac{\beta}{n\sin(\theta_0+\theta)} & \text{if } \theta_0 + \theta \in [\theta_1, \theta_2], \\ \dfrac{1}{n\sin(\theta_0+\theta)} & \text{if } \theta_0 + \theta \in [\theta_2, \theta_3], \\ \dfrac{\beta+1}{n\sin(\theta_0+\theta)} - \dfrac{\alpha}{n\cos(\theta_0+\theta)} & \text{if } \theta_0 + \theta \in [\theta_3, \theta_4], \end{cases}$$

where $0 < \theta_1 < \theta_2 < \theta_3 < \theta_4 < \pi/2$ are the angles between the horizontal axis and the particular geodesics passing through the corners of c_{ij}^n (see Figure 4). Note that this formula is still satisfied when $\theta_1 = 0$ or $\theta_4 = \pi/2$. In such cases, either α or β vanishes. The claim (26) is then proved.

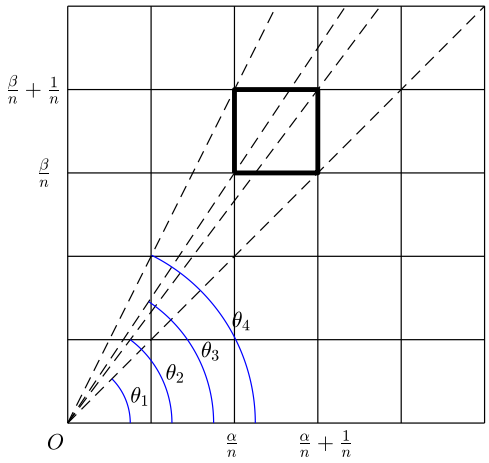


FIGURE 4. Particular geodesics issued from O and meeting the square with bold boundary.

Note that the rotation R_θ can be applied to γ until the geodesic ray γ_θ meets a new corner. It is then easy to see that

$$-\theta_- \le \frac{C}{n}, \qquad \theta_+ \le \frac{C}{n}, \tag{27}$$

for some $C > 0$ neither depending on n nor on γ.

According to (26), we have

$$M_{\gamma_\theta}(\omega) = \frac{X_1}{\sin(\theta_0+\theta)} + \frac{X_2}{\cos(\theta_0+\theta)}, \tag{28}$$

with

$$X_1 = \frac{1}{n}\sum_{(i,j)\in I'(\gamma)} a_{ij} M_{\gamma_\theta}(c_{ij}^n) \quad \text{and} \quad X_2 = \frac{1}{n}\sum_{(i,j)\in I'(\gamma)} b_{ij} M_{\gamma_\theta}(c_{ij}^n),$$

for $\theta \in J$. Moreover, we have

$$\max(|X_1|, |X_2|) \le \frac{1}{n}\,\#I'(\gamma) n\,\frac{\sqrt{2}}{nT} \le 3$$

because $\max(|a_{ij}|, |b_{ij}|) \le n$, $M_{\gamma_\theta}(c_{ij}^n) \le \sqrt{2}/nT$ and $\#I'(\gamma) \le 2n$, the latter inequality following from the obvious observation that, when γ leaves a square c_{ij}^n, then the next square that γ enters must be of the form $c_{i'j'}$ with $i' = i+1$ or $j' = j+1$.

We are now in a position to establish (24). It is obvious if $\min_J f$ is reached at θ_- or θ_+. Let us then assume that f reaches its minimum in (θ_-, θ_+); without loss of generality, we assume that the minimum is reached at $\theta = 0$. In particular, $\frac{d}{d\theta}\big|_{\theta=0} M_{\gamma_\theta}(\omega) = 0$ which, using (28), yields

$$\frac{\cos(\theta_0)}{\sin^2(\theta_0)}\,X_1 = \frac{\sin(\theta_0)}{\cos^2(\theta_0)}\,X_2.$$

From (28), we also have $f(0) = {}^{X_1}/_{\sin\theta_0} + {}^{X_2}/_{\cos\theta_0}$ and thus $X_1 = f(0)\sin^3(\theta_0)$ and $X_2 = f(0)\cos^3(\theta_0)$, so that

$$f(\theta) = M_{\gamma_\theta}(\omega) = f(0)\Big(\frac{\sin^3(\theta_0)}{\sin(\theta_0+\theta)} + \frac{\cos^3(\theta_0)}{\cos(\theta_0+\theta)}\Big).$$

Now,

$$f(\theta) - f(0) = f(0)\Big(\sin^3\theta_0\Big(\frac{1}{\sin(\theta_0+\theta)} - \frac{1}{\sin\theta_0}\Big) + \cos^3\theta_0\Big(\frac{1}{\cos(\theta_0+\theta)} - \frac{1}{\cos\theta_0}\Big)\Big).$$

This computation shows that $\theta_0 + \theta_+ \neq \pi/2$ and $\theta_0 + \theta_- \neq 0$. Otherwise, since $\theta_0 \notin \{0, \pi/2\}$, f would be not bounded on J when θ tends to $\theta_0+\theta_+$ or $\theta_0+\theta_-$, which is not true by definition of f. This implies that

$$\arcsin({}^1/_{nT}) \le \theta_0 + \theta_- < \theta_0 + \theta_+ \le \frac{\pi}{2} - \arcsin({}^1/_{nT}). \tag{29}$$

Indeed, set $\gamma_- = \gamma_{\theta_0+\theta_-}$. We know that $\gamma_-([0,T])$ contains at least two corners and is not horizontal (since $\theta_0+\theta_- > 0$). This means that its second coordinate (in $\mathbb{R}^2$) increases at least of $1/n$ between 0 and T and hence $T\sin(\theta_0+\theta_-) \ge 1/n$. A similar argument gives the right-hand inequality in (29).

From now on, we let n tend to $+\infty$. In particular, θ_0 depends on n. Let us prove that

$$f(\theta_-) - \min_J f = f(\theta_-) - f(0) \leq \frac{C}{n}. \tag{30}$$

A similar argument will show that $f(\theta_+) - \min_J f \leq C/n$. These two estimates imply (24) and hence complete the proof of the item b. It thus remains to prove (30). By the mean value theorem, there exists $\tilde\theta \in (0, |\theta_-|)$ such that

$$\frac{1}{\sin(\theta_0+\theta_-)} - \frac{1}{\sin\theta_0} = \frac{\cos(\theta_0-\tilde\theta)}{\sin^2(\theta_0-\tilde\theta)}\,\theta_-.$$

Since $0 \leq \tilde\theta \leq |\theta_-|$, using that $|\sin(\theta_0)\cos(\theta_0-\tilde\theta)| \leq 1$, we get from (27) that

$$\sin^3(\theta_0)\Big|\frac{1}{\sin(\theta_0+\theta_-)} - \frac{1}{\sin\theta_0}\Big| \leq \frac{C}{n}\Big(\frac{\sin(\theta_0)}{\sin(\theta_0-\tilde\theta)}\Big)^2.$$

Now, by the mean value theorem,

$$\frac{\sin(\theta_0)}{\sin(\theta_0-\tilde\theta)} = 1 + \frac{\sin(\theta_0)-\sin(\theta_0-\tilde\theta)}{\sin(\theta_0-\tilde\theta)} \leq 1 + \frac{\mathrm{O}(1/n)}{\sin(\theta_0-\tilde\theta)}.$$

Using (29), there exists $C' > 0$ neither depending on γ nor on n such that $\mathrm{O}(1/n)/\sin(\theta_0-\tilde\theta) = \mathrm{O}(1)$. This implies that

$$\sin^3(\theta_0)\Big(\frac{1}{\sin(\theta_0-\tilde\theta)} - \frac{1}{\sin\theta_0}\Big) = \mathrm{O}(1/n).$$

Following the same reasoning, we prove that

$$\cos^3(\theta_0)\Big(\frac{1}{\cos(\theta_0-\tilde\theta)} - \frac{1}{\cos\theta_0}\Big) = \mathrm{O}(1/n).$$

Actually, this last estimate is much easier since $|\cos(\theta_0)| \leq C\cos(\theta_0+\theta_-)$. This leads to (30).

3.2.3. *Proof of Lemma 6.* Let $(i_1,j_1),(i_2,j_2) \in \{1,\dots,n\}^2$ be such that the two square cells $c_{i_1j_1}$ and $c_{i_2j_2}$ of $\mathcal{G}^n$ are distinct and let x_1 and x_2 be two distinct corners of the grid $\mathcal{G}^n$. Any geodesic ray such that

$$\gamma(0) \in c_{i_1j_1}, \quad \gamma(T) \in c_{i_2j_2} \quad \text{and} \quad x_i \in \gamma([0,T]),\ i=1,2,$$

will be denoted by $\gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}$. The set of all geodesic rays $\gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}$ is denoted by $\Gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}$. Given any $T \in (0,1)$, this set is nonempty as soon as n is large enough. In order to obtain a finite set, we make the following observation. Considering a ray $\gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}$, any other ray γ passing through x_1 and x_2 and such that $\gamma(0) \in c_{i_1j_1}$ and $\gamma(T) \in c_{i_2j_2}$ is obtained from $\gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}$ by a time translation. This creates an equivalence relation and we define $\hat\Gamma$ as the quotient of Γ by this equivalence relation. Any element $\hat\gamma$ of $\hat\Gamma$ is uniquely determined by a choice of distinct pairs $(i_1,j_1),(i_2,j_2) \in \{1,\dots,n\}^2$ and of distinct $x_1,x_2 \in \mathcal{G}^n$. Therefore, by construction, we have $\#\hat\Gamma = \mathrm{O}(n^8)$.

Let $\hat\gamma \in \hat\Gamma$. If $\hat\gamma$ is neither horizontal nor vertical then $m^T_{\hat\gamma}(\omega^n_\varepsilon)$ can be expressed by (10) and, using (9) and applying the large deviation result established in Proposition 1 in Section 3.3, we get

$$\mathbb{P}(m^T_{\hat\gamma}(\omega^n_\varepsilon) \leq \varepsilon - \delta) \leq C_{\varepsilon,\delta}\, e^{-m\delta^2 T/(T+1)\sqrt{2}},$$

where m is the number of square cells having a nontrivial intersection with $\hat\gamma([0,T])$. Indeed, $m \leq n$ because $T < 1$ and thus, by (9),

$$\lambda_i = \frac{t^n_{ij}(\gamma)}{T} \leq \frac{T+1}{T}\,\frac{\sqrt{2}}{n} \leq \frac{c}{m} \qquad \text{with } c = \frac{T+1}{T}\,\sqrt{2} > 1.$$

If $\hat\gamma$ is horizontal or vertical, the argument is more complicated. We consider $\widetilde{m}^T_{\hat\gamma}(\omega^n_\varepsilon)$ defined by (8) in Section 3.1. Obviously, we have

$$\widetilde{m}^T_{\hat\gamma}(\omega^n_\varepsilon) = \inf \mathcal{L},$$

where $\mathcal{L}$ is the set of all possible limits (closure points) of $m^T_{\gamma'_k}(\omega^n_\varepsilon)$ as $k \to +\infty$ over all sequences $(\gamma'_k)_{k\in\mathbb{N}}$ of geodesic rays converging to $\hat\gamma$ such that, for every $k \in \mathbb{N}$, γ'_k is obtained by rotating $\hat\gamma$ around a corner of the grid through which $\hat\gamma$ passes, with angle $\pm 1/k$. There are at most $\lfloor n/T \rfloor + 1$ corners belonging to $\hat\gamma([0,T])$ and thus $2(\lfloor n/T \rfloor + 1)$ possible limits of $m^T_{\gamma'_k}(\omega^n_\varepsilon)$ by considering positive and negative rotations. Hence $\#\mathcal{L} \leq 2\lfloor n/T \rfloor + 2$.

We claim that any $m_\infty \in \mathcal{L}$ can be expressed by (10) so that we can still apply Proposition 1. Indeed, assume for instance that γ'_k is obtained from $\hat\gamma$ by a rotation of angle $1/k$ around a corner c of the grid through which $\hat\gamma$ passes. Then, it is easy to see that

$$m_\infty = \sum_{i,j=1}^{n} \frac{t_{ij}^n(\hat\gamma)}{T}\, X_{ij}^n,$$

where $t_{ij}^n(\hat\gamma)$ is the time spent by $\hat\gamma$ in c_{ij}^n if γ'_k crosses c_{ij}^n (it does not depend on k for k large enough) and 0 otherwise. More precisely, by Proposition 1, we have $\mathbb{P}(m_\infty \le \varepsilon - \delta) \le C_{\varepsilon,\delta}\, e^{-m\delta^2 T/(T+1)\sqrt{2}}$. Hence, using that $\#\mathcal{L} \le 2\lfloor n/T \rfloor + 2$ and modifying the constant $C_{\varepsilon,\delta}$, we obtain

$$\mathbb{P}\left(\widetilde{m}_{\hat\gamma}^T(\omega_\varepsilon^n) \le \varepsilon - \delta\right) \le \sum_{m_\infty \in \mathcal{L}} \mathbb{P}\left(m_\infty \le \varepsilon - \delta\right) \le C_{\varepsilon,\delta}\, n\, e^{-m\delta^2 T/(T+1)\sqrt{2}}.$$

In all cases, let us estimate m, the number of square cells met by $\hat\gamma$. Since the diagonal length of each square cell c_{ij}^n is $\sqrt{2}/n$, the length of $\hat\gamma([0,T])$ (which is equal to T because the speed of the geodesic is 1) is bounded above by $m\sqrt{2}/n$. Therefore $m \ge nT/\sqrt{2}$.

We have therefore proved that, for every geodesic ray $\hat\gamma \in \hat\Gamma$ (horizontal or vertical or not),

$$\mathbb{P}\left(\widetilde{m}_{\hat\gamma}^T(\omega_\varepsilon^n) \le \varepsilon - \delta\right) \le C_{\varepsilon,\delta}\, n\, e^{-nT^2\delta^2/2(T+1)}. \tag{31}$$

Now, let $\gamma \in \Gamma^2$. Assuming that n is large enough, there exist distinct pairs $(i_1,j_1),(i_2,j_2) \in \{1,\dots,n\}^2$ and two distinct corners x_1, x_2 in $\mathcal{G}^n$ such that $\gamma(0) \in c_{i_1j_1}$, $\gamma(T) \in c_{i_2j_2}$ and $x_1, x_2 \in \gamma([0,T])$. Since γ and $\gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}$ are in the same equivalence class, and since the diagonal length of a square cell of $\mathcal{G}^n$ is $\sqrt{2}/n$, we have

$$\left|m_\gamma^T(\omega_\varepsilon^n) - m^T_{\gamma_{c_{i_1j_1},c_{i_2j_2},x_1,x_2}}(\omega_\varepsilon^n)\right| \le \frac{2\sqrt{2}}{n}.$$

We infer that

$$\inf_{\hat\gamma\in\hat\Gamma} m_{\hat\gamma}^T(\omega_\varepsilon^n) = \ell_{\Gamma^2}^T(\omega_\varepsilon^n) + \mathrm{O}(1/n).$$

Therefore, recalling that $\widetilde{m}_\gamma^T(\omega_\varepsilon^n) \le m_\gamma^T(\omega_\varepsilon^n)$ (see (8)), there exists $C > 0$ (not depending on n) such that, if n is large enough, then

$$\begin{aligned}\mathbb{P}\left(\ell_{\Gamma^2}^T(\omega_\varepsilon^n) \le \varepsilon - \delta\right) &\le \mathbb{P}\big(\inf_{\gamma\in\hat\Gamma} m_\gamma^T(\omega_\varepsilon^n) \le \varepsilon - 2\delta/3\big) \\ &\le \sum_{\hat\gamma\in\hat\Gamma} \mathbb{P}\left(m_{\hat\gamma}^T(\omega_\varepsilon^n) \le \varepsilon - 2\delta/3\right) \le Cn^9 e^{-2nT^2\delta^2/9(T+1)}\end{aligned}$$

because $\#\hat\Gamma = \mathrm{O}(n^8)$. The conclusion follows.

3.3. **Large deviations for triangular arrays of i.i.d. Bernoulli variables.**

Proposition 1. *Let $m \ge 3$ be an integer and let $(\lambda_1,\dots,\lambda_m)$ be a m-tuple of nonnegative real numbers satisfying $\sum_{i=1}^m \lambda_i = 1$ and $\lambda_i \le c/m$ for every i, for some $c > 1$. We set $Y_m = \sum_{i=1}^m \lambda_i X_i$, where $(X_1,\dots,X_m)$ is a m-tuple of i.i.d. Bernoulli variables with expectation $\varepsilon \in [0,1]$. Then, for every $\delta > 0$, there exists $C_{\varepsilon,\delta} > 0$ such that*

$$\mathbb{P}(|Y_m - \varepsilon| \ge \delta) \le C_{\varepsilon,\delta}\, e^{-\delta^2 m/c}. \tag{32}$$

Remark 2. The assumptions that $c > 1$ and $m \ge 3$ are not restrictive. They allow us to provide a sharp estimate of the right-hand side of (32), by solving an auxiliary optimization problem.

Proof of Proposition 1. We have

$$\mathbb{P}(|Y_m - \varepsilon| \ge \delta) = \mathbb{P}\left(Y_m \ge \varepsilon + \delta\right) + \mathbb{P}\left(-Y_m \ge -\varepsilon + \delta\right)$$

and it suffices to prove that

$$\mathbb{P}(Y_m \ge \varepsilon + \delta) \le C_{\varepsilon,\delta}\, e^{-\delta^2 m/c}$$

because the estimate on $\mathbb{P}\left(-Y_m \ge -\varepsilon + \delta\right)$ is obtained similarly.

Let $s > 0$ to be chosen later. By the Markov inequality, we have

$$\mathbb{P}(Y_m \ge \varepsilon + \delta) = \mathbb{P}\big(e^{sm(Y_m - \varepsilon - \delta)} \ge 1\big) \le \mathbb{E}\big(e^{sm(Y_m-\varepsilon-\delta)}\big) = e^{-sm(\delta+\varepsilon)}\mathbb{E}\big(e^{smY_m}\big).$$

Using and the independence of the Bernoulli variables X_i (whose expectation is ε), we infer that

$$\mathbb{P}(Y_m \geq \varepsilon + \delta) \leq e^{-sm(\delta+\varepsilon)} \prod_{i=1}^{m} \mathbb{E}\left(e^{sm\lambda_i X_i}\right) = e^{-sm(\delta+\varepsilon)} \prod_{i=1}^{m} \left(1 - \varepsilon + \varepsilon e^{sm\lambda_i}\right) \\ = e^{-sm(\delta+\varepsilon)} F(\lambda_1, \ldots, \lambda_m) \leq e^{-sm(\delta+\varepsilon)} \max_{\Sigma_m} F, \tag{33}$$

where

$$F(\lambda_1, \ldots, \lambda_m) = \prod_{i=1}^{m} \left(1 - \varepsilon + \varepsilon e^{sm\lambda_i}\right)$$

and

$$\Sigma_m = \left\{(\lambda_1, \cdots, \lambda_m) \in [0, c/m]^m \mid \textstyle\sum_{i=1}^m \lambda_i = 1\right\}.$$

Lemma 7. *We have*

$$\max_{\Sigma_m} F = (1 - \varepsilon + \varepsilon e^{sc})^{\lfloor m/c \rfloor} \left(1 - \varepsilon + \varepsilon e^{s(m - c\lfloor m/c \rfloor)}\right),$$

where $\lfloor \cdot \rfloor$ denotes the floor function.

Proof of Lemma 7. Let $(a_1, \ldots, a_m) \in \Sigma_m$ be a point at which the continuous function F reaches its maximum over the compact set Σ_m. Let $(j, k) \in \{1, \ldots, m\}^2$ be such that $j \neq k$. We define the function α on $\mathbb{R}$ by

$$\alpha(u) = F(a_1, \ldots, a_j + u, \ldots, a_k - u, \ldots, a_m).$$

Setting $I_{j,k} = [\max(-a_j, a_k - c/m), \min(a_k, c/m - a_j)]$, for every $u \in I_{j,k}$ (i.e.$0 \leq a_j + u \leq c/m$ and $0 \leq a_k - u \leq c/m$, we have $(a_1, \ldots, a_j + u, \ldots, a_k - u, \ldots, a_m) \in \Sigma_m$. Note that $0 \in I_{j,k}$ and that, since $(a_1, \ldots, a_m)$ is a maximizer of F, we have $\alpha(u) \leq \alpha(0)$ for every $u \in I_{j,k}$. We have two possible cases:

(i) $a_j = 0$ or $a_j = c/m$ or $a_k = 0$ or $a_k = c/m$;
(ii) $0 < a_j < c/m$ and $0 < a_m < c/m$.

In the case (ii), we must have $\alpha'(0) = 0$, and since, by computing this derivative, we have

$$\alpha'(0) = sm\varepsilon(1 - \varepsilon)\left(e^{sma_j} - e^{sma_k}\right) \prod_{\substack{1 \leq i \leq m \\ i \neq j,\ i \neq k}} (1 - \varepsilon + \varepsilon e^{sma_i}),$$

it follows that $a_j = a_k$.

Since the pair (j, k) of distinct integers was arbitrary, we conclude that there exists $\bar{\lambda} \in (0, c/m)$ such that $a_j \in \{0, \bar{\lambda}, c/m\}$ for every $j \in \{1, \ldots, m\}$. Let J be the set of indices such that $a_j = \bar{\lambda}$ for every $j \in J$. Denote by F_J the restriction of F to the set of all $(\lambda_1, \ldots, \lambda_m) \in \Sigma_m$ such that $\lambda_i = a_i$ for every $i \notin J$. Observing that F_J is the product of separate variables positive strictly convex functions, its Hessian $d^2F_J(a_1, \ldots, a_m)$ must be positive definite. But, by maximality of $(a_1, \ldots, a_m)$ and since $\sum_{i=1}^m a_i = 1$, $d^2F_J(a_1, \ldots, a_m)$ has at most one positive eigenvalue. Therefore J contains at most one element.

Noting that F is invariant under permutations, we infer that, performing a permutation of variables if necessary, there exists $N \in \{1, \ldots, m-2\}$ such that

$$a_i = \frac{c}{m} \quad \forall i \in \{1, \ldots, N\} \qquad \text{and} \qquad a_j = 0 \quad \forall j \in \{N+1, \ldots, m-1\}$$

and $a_m = 1 - Nc/m$ because $\sum_{i=1}^m a_i = 1$. Since $a_m \geq 0$, we must have

$$N \leq \min(\lfloor m/c \rfloor, m - 2) = \lfloor m/c \rfloor$$

because $c > 1$ and $m \geq 3$. Therefore

$$\max_{\Sigma_m} F = e^{\varphi(N)} \quad \text{with} \quad \varphi(x) = x \ln(1 - \varepsilon + \varepsilon e^{sc}) + \ln(1 - \varepsilon + \varepsilon e^{s(m - cx)}).$$

It remains to determine the best possible integer $N \in \{1, \ldots, \lfloor m/c \rfloor\}$. We have

$$\varphi'(N) = \ln(1 - \varepsilon + \varepsilon e^{sc}) - \frac{sc\varepsilon e^{s(m-cN)}}{1 - \varepsilon + \varepsilon e^{s(m-cN)}} \\ \geq \frac{sc\varepsilon(1 - \varepsilon)}{1 - \varepsilon + \varepsilon e^{s(m-cN)}}(1 - e^{s(m-cN)}) > 0,$$

where we have used that $1 - \varepsilon + \varepsilon e^{s(m-cN)} > 1$ and $\ln(1 - \varepsilon + \varepsilon e^{sc}) \geq \varepsilon sc$ by concavity of the logarithm. Hence, the best integer N is $N = \lfloor m/c \rfloor$. The result follows. □

Using (33), Lemma 7 and the inequalities $m - c\lfloor m/c \rfloor \le c$ and $1 < 1 - \varepsilon + \varepsilon e^{sc} \le e^{sc}$, we obtain

$$\begin{aligned}\mathbb{P}(Y_m \ge \varepsilon + \delta) &\le e^{-sm(\delta+\varepsilon)}(1-\varepsilon+\varepsilon e^{sc})^{\lfloor m/c \rfloor + 1} \le e^{-sm(\delta+\varepsilon)}(1-\varepsilon+\varepsilon e^{sc})^{(m/c)+1} \\ &\le (1-\varepsilon+\varepsilon e^{sc}) e^{-m(cs(\varepsilon+\delta)-\ln(1-\varepsilon+\varepsilon e^{cs}))/c} \le e^{sc} e^{-m\varphi_{\varepsilon,\delta}(cs)/c},\end{aligned} \tag{34}$$

where

$$\varphi_{\varepsilon,\delta}(u) = u(\varepsilon+\delta) - \ln(1-\varepsilon+\varepsilon e^u).$$

In order to choose adequately the parameter s, we will use the following result.

Lemma 8. *For every $\delta > 0$, we have*

$$\inf_{u>0} \varphi_{\varepsilon,\delta}(u) < \delta^2 < \sup_{u>0} \varphi_{\varepsilon,\delta}(u).$$

Moreover, there exists $\hat{u}_{\varepsilon,\delta} \ge 0$ such that $\varphi_{\varepsilon,\delta}(\hat{u}_{\varepsilon,\delta}) = \delta^2$, and $\hat{u}_{\varepsilon,\delta} \le U_{\varepsilon,\delta}$, where

$$U_{\varepsilon,\delta} = \begin{cases} \delta^2/(\varepsilon+\delta-1) & \text{if } \delta > 1-\varepsilon, \\ \ln\big((1-\varepsilon)e^{\delta^2}/(1-\varepsilon e^{\delta^2})\big) & \text{if } \delta = 1-\varepsilon, \\ \ln\big((1-\varepsilon)(\varepsilon+\delta)/\varepsilon(1-\varepsilon-\delta)\big) & \text{if } \delta < 1-\varepsilon. \end{cases}$$

Proof of Lemma 8. We distinguish between several cases:

- If $\delta > 1-\varepsilon$, using that $\ln(1-\varepsilon+\varepsilon e^u) \le u$ for every $u \ge 0$, it follows that $\varphi_{\varepsilon,\delta}(u) \ge (\varepsilon+\delta-1)u$ and hence
$$\sup_{u>0} \varphi_{\varepsilon,\delta}(u) \ge \varphi_{\varepsilon,\delta}\Big(\frac{\delta^2}{\varepsilon+\delta-1}\Big) \ge \delta^2.$$
Moreover, since $\varphi_{\varepsilon,\delta}(0) = 0$, by continuity of $\varphi_{\varepsilon,\delta}$, there exists $\hat{u}_{\varepsilon,\delta} \in (0, \delta^2/\varepsilon+\delta-1)$ such that $\varphi_{\varepsilon,\delta}(\hat{u}_{\varepsilon,\delta}) = \delta^2$.
- If $\delta = 1-\varepsilon$, since $\varphi_{\varepsilon,\delta}$ is monotone increasing and $\varphi_{\varepsilon,\delta}(u) \le \lim_{u\to+\infty} \varphi_{\varepsilon,\delta}(u) = -\ln\varepsilon$, we obtain
$$\sup_{u>0} \varphi_{\varepsilon,\delta}(u) = -\ln(1-\delta) \ge \delta \ge \delta^2.$$
Moreover, noting that $\varphi_{\varepsilon,\delta}(u) = \ln\big(e^u/(1-\varepsilon+\varepsilon e^u)\big)$, $\hat{u}_{\varepsilon,\delta}$ can be computed and we obtain the formula stated in Lemma 8.
- If $\delta < 1-\varepsilon$, we set $u_{\varepsilon,\delta} = \ln\big((\varepsilon+\delta)(1-\varepsilon)/\varepsilon(1-\varepsilon-\delta)\big)$. A straightforward study shows that $\varphi_{\varepsilon,\delta}$ is monotone increasing on $[0, u_{\varepsilon,\delta}]$ and monotone decreasing on $[u_{\varepsilon,\delta}, +\infty)$ and then

$$\begin{aligned}\sup_{u>0} \varphi_{\varepsilon,\delta}(u) &= (\varepsilon+\delta)\ln\Big(\frac{(\varepsilon+\delta)(1-\varepsilon)}{\varepsilon(1-\varepsilon-\delta)}\Big) - \ln\Big(\frac{1-\varepsilon}{1-\varepsilon-\delta}\Big) \\ &= (\varepsilon+\delta)\ln\Big(\frac{\varepsilon+\delta}{\varepsilon}\Big) + (\varepsilon+\delta-1)\ln\Big(\frac{1-\varepsilon}{1-\varepsilon-\delta}\Big).\end{aligned}$$

Setting $a = \varepsilon+\delta \in (\varepsilon, 1)$, we get that

$$\sup_{u>0} \varphi_{\varepsilon,\delta}(u) = f(a) \quad \text{with} \quad f(a) = a\ln\Big(\frac{a}{\varepsilon}\Big) + (1-a)\ln\Big(\frac{1-a}{1-\varepsilon}\Big).$$

We compute

$$f'(a) = \ln\Big(\frac{a}{\varepsilon}\Big) - \ln\Big(\frac{1-a}{1-\varepsilon}\Big) \quad \text{and} \quad f''(a) = \frac{1}{a(1-a)}.$$

Note that $f''(a) \ge 4$ for every $a \in (0,1)$. Integrating two times this inequality and using that $f(\varepsilon) = f'(\varepsilon) = 0$, we get $f(a) \ge 2\delta^2$ and therefore

$$\sup_{u>0} \varphi_{\varepsilon,\delta}(u) \ge 2\delta^2 > \delta^2.$$

The lemma is proved. □

We choose s such that $cs = \hat{u}_{\varepsilon,\delta}$, where $\hat{u}_{\varepsilon,\delta}$ is given by Lemma 8, and we set $C_{\varepsilon,\delta} = e^{\hat{u}_{\varepsilon,\delta}}$. The inequality (34) yields $\mathbb{P}(Y_m \ge \varepsilon+\delta) \le C_{\varepsilon,\delta}\, e^{-m\delta^2/c}$. Proposition 1 is proved. □

3.4. **Proof of Corollary 3.** It is proved in [6] that $\ell^T(\omega) \leq |\omega|$ for every Riemann integrable subset ω of $M = \mathbb{T}^2$. Let $\delta > 0$ be arbitrary. Let us apply Theorem 2 by changing the parameter ε of the Bernoulli law to $\varepsilon' = \varepsilon - \delta$. Noting that the random variable $|\omega^n_{\varepsilon'}|$ follows the law $(1/n^2)B(n^2, \varepsilon')$, we obtain

$$\lim_{n\to+\infty} \mathbb{P}(|\omega^n_{\varepsilon'}| \leq \varepsilon' + \delta) = \lim_{n\to+\infty} \mathbb{P}(|\omega^n_{\varepsilon'}| \leq \varepsilon) = 1$$

and

$$\lim_{n\to+\infty} \mathbb{P}(\ell^T(\omega^n_{\varepsilon'}) \geq \varepsilon' - \delta) = \lim_{n\to+\infty} \mathbb{P}(\ell^T(\omega^n_{\varepsilon'}) \geq \varepsilon - 2\delta) = 1.$$

As a consequence,

$$\lim_{n\to+\infty} \mathbb{P}(\ell^T(\omega^n_{\varepsilon'}) \leq \varepsilon - 2\delta \text{ or } |\omega^n_{\varepsilon'}| \geq \varepsilon) \leq \lim_{n\to+\infty} \mathbb{P}(\ell^T(\omega^n_{\varepsilon'}) \leq \varepsilon - 2\delta) + \lim_{n\to+\infty} \mathbb{P}(|\omega^n_{\varepsilon'}| \geq \varepsilon) = 0$$

and therefore,

$$\lim_{n\to+\infty} \mathbb{P}\left(\ell^T(\omega^n_{\varepsilon'}) \geq \varepsilon - 2\delta \text{ and } |\omega^n_{\varepsilon'}| \leq \varepsilon\right) = 1,$$

which yields the existence of a domain ω that is a union of subsquares of a grid $\mathcal{G}^n$ for some $n \in \mathbb{N}^*$ such that $\ell^T(\omega) \geq \varepsilon - 2\delta$ and $|\omega| \leq \varepsilon$. We then infer that $\varepsilon - 2\delta \leq \sup_{|\omega|\leq\varepsilon} \ell^T(\omega) = \varepsilon$ and we conclude since δ has been chosen arbitrarily. Corollary 3 is proved.

Appendix. Some properties of the functional ℓ^T

Recall that, given any $T > 0$ and any Lebesgue measurable subset ω of M, denoting by χ_ω the characteristic function of ω, we have defined

$$\ell^T(\omega) = \inf_{\gamma\in\Gamma} \frac{1}{T} \int_0^T \chi_\omega(\gamma(t))\, dt.$$

The functional ℓ^T can be extended by replacing χ_ω by any measurable function a on M. It can even be extended further: any geodesic ray $\gamma \in \Gamma$ is the projection onto M of a geodesic curve on S^*M, that is, $\gamma(t) = \pi \circ \varphi_t(z)$ for some $z \in S^*M$. Here, we denote by $(\varphi_t)_{t\in\mathbb{R}}$ the Riemannian geodesic flow, where, for every $t \in \mathbb{R}$, φ_t is a symplectomorphism on (T^*M, ω) which preserves S^*M. Now, given any bounded measurable function a on (S^*M, μ_L) and given any $T > 0$, we define

$$\ell^T(a) = \inf_{z\in S^*M} \frac{1}{T} \int_0^T a \circ \varphi_t(z)\, dt = \inf_{z\in S^*M} \bar{a}_T(z),$$

where $\bar{a}_T(z) = \frac{1}{T}\int_0^T a \circ \varphi_t(z)\, dt$ and where the unit cotangent bundle S^*M is endowed with the Liouville measure μ_L. Note that $\ell^T(a) = \ell^T(a \circ \varphi_t)$, i.e.$\ell^T$ is invariant under the geodesic flow.

It can also be noted that for a fixed the function $T \mapsto T\ell^T(a)$ is superadditive. Of course, we recover the initial definition of ℓ^T by pushforward to M under the canonical projection $\pi : S^*M \to M$: given any bounded measurable function f on (M, dx_g), we have

$$(\pi_*\ell^T)(f) = \ell^T(\pi^* f) = \ell^T(f \circ \pi) = \inf_{z\in S^*M} \frac{1}{T} \int_0^T f \circ \pi \circ \varphi_t(z)\, dt = \inf_{\gamma\in\Gamma} \frac{1}{T} \int_0^T f(\gamma(t))\, dt$$

that we simply denote by $\ell^T(f)$. When $f = \chi_\omega$, we recover $\ell^T(\omega)$.

Remark 3. Setting $a_t = a \circ \varphi_t$, and assuming that $a \in C^\infty(S^*M)$ is the principal symbol of a pseudo-differential operator $A \in \Psi^0(M)$ (of order 0), that is, $a = \sigma_P(A)$, we have, by the Egorov theorem (see [10]),

$$a_t = a \circ \varphi_t = \sigma_P(A_t) \qquad \text{with} \qquad A_t = e^{-it\sqrt{\Delta}} A e^{it\sqrt{\Delta}}$$

where $\sigma_P(\cdot)$ is the principal symbol. Accordingly, we have $\bar{a}_T = \sigma_P(\bar{A}_T)$ with

$$\bar{A}_T = \frac{1}{T}\int_0^T A_t\, dt = \frac{1}{T}\int_0^T e^{-it\sqrt{\Delta}} A e^{it\sqrt{\Delta}}\, dt.$$

We provide hereafter a microlocal interpretation of the functionals ℓ^T and we give a relationship with the wave observability constant.

Microlocal interpretation of ℓ^T and of the wave observability constant. Let f_T be such that $\hat{f}_T(t) = (1/T)\chi_{[0,T]}(t)$, i.e.

$$f_T(t) = \frac{1}{2\pi}\, i e^{iTt/2}\,\mathrm{sinc}(Tt/2).$$

Note that $\int_{\mathbb{R}} \hat{f}_T = 1$, i.e.equivalently, $f_T(0) = 1$. We denote by X the Hamiltonian vector field on S^*M of the geodesic flow (we have $e^{tX} = \varphi_t$ for every $t \in \mathbb{R}$), and we define the selfadjoint operator $S = X/i$. Using that $a \circ e^{tX} = (e^{tX})^* a = e^{tL_X} a = e^{itS} a$, we get

$$\ell^T(a) = \inf_{z\in S^*M} \frac{1}{T}\int_0^T a\circ e^{tX}(z)\,dt = \inf_{z\in S^*M} \int_{\mathbb{R}} \hat{f}_T(t) e^{itS} a\,dt\ (z) = \inf f_T(S)a.$$

Besides, setting $A = \mathrm{Op}(a)$ (where Op is a quantization), we have

$$\bar{A}_T(a) = \frac{1}{T}\int_0^T e^{-it\sqrt{\triangle}}\mathrm{Op}(a)e^{it\sqrt{\triangle}}\,dt = \int_{\mathbb{R}} \hat{f}_T(t) e^{-it\sqrt{\triangle}}\mathrm{Op}(a)e^{it\sqrt{\triangle}}\,dt$$
$$= A_{f_T} = \sum_{\lambda,\mu} f_T(\lambda-\mu) P_\lambda A P_\mu,$$

where P_λ is the projection onto the eigenspace corresponding to the eigenvalue λ of $\sqrt{\triangle}$, i.e.$\sqrt{\triangle} = \sum_{\lambda\in\mathrm{Spec}(\sqrt{\triangle})} \lambda P_\lambda$. Restricting to half-waves, the wave observability constant is therefore given (see [7]) by

$$C_T(a) = \inf_{\|y\|=1} \langle \bar{A}_T(a)y, y\rangle = \inf_{\|y\|=1} \langle A_{f_T} y, y\rangle.$$

Note that, for every $y = \sum_\lambda P_\lambda y = \sum_\lambda y_\lambda \phi_\lambda \in L^2(M)$, where ϕ_λ is an eigenfunction of norm 1 associated with λ, for every smooth function a on M (i.e.A is the operator of multiplication by a), we have

$$\langle A_{f_T} y, y\rangle = \sum_{\lambda,\mu} f_T(\lambda-\mu)\langle AP_\lambda y, P_\mu y\rangle = \sum_{\lambda,\mu} f_T(\lambda-\mu)\int_M a\, P_\lambda y\,\overline{P_\mu y}$$
$$= \sum_{\lambda,\mu} f_T(\lambda-\mu) y_\lambda \bar{y}_\mu \int_M a\phi_\lambda\phi_\mu$$

and we thus recover the expression of $C_T(a)$ by series expansion (see [7]).

Note also that, as said before, the principal symbol of $A_{f_T} = \bar{A}_T(a)$ is

$$\sigma_P(A_{f_T}) = \sigma_P(\bar{A}_T(a)) = a_{f_T} = \int_{\mathbb{R}} \hat{f}_T(t) a\circ e^{tX}\,dt = f_T(S)a$$

and thus

$$\ell^T(a) = \inf \sigma_P(\bar{A}_T(a)).$$

Semi-continuity properties of ℓ^T. Note the obvious fact that if a and b are functions such that $a \le b$ and for which the following quantities make sense, then $\ell^T(a) \le \ell^T(b)$. In other words, the functional ℓ^T is nondecreasing.

Lemma 9. *Let ω be a subset of M, let $T > 0$ be arbitrary and let $(h_k)_{k\in\mathbb{N}^*}$ be a uniformly bounded sequence of Borel measurable functions on M. If h_k converges pointwise to χ_ω, then*

$$\limsup_{k\to+\infty} \ell^T(h_k) \le \ell^T(\omega)$$

and if moreover $\chi_\omega \le h_k$ for every $k \in \mathbb{N}^*$, then

$$\ell^T(\omega) = \lim_{k\to+\infty} \ell^T(h_k) = \inf_{k\in\mathbb{N}^*} \ell^T(h_k).$$

Proof. Let $\gamma \in \Gamma$ be arbitrary. By pointwise convergence, we have $h_k(\gamma(t)) \to \chi_\omega(\gamma(t))$ for every $t \in [0,T]$, and it follows from the dominated convergence theorem that $\ell^T(h_k) \le \frac{1}{T}\int_0^T h_k(\gamma(t))\,dt \to \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\,dt$, and thus $\limsup_{k\to+\infty} \ell^T(h_k) \le \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\,dt$. Since this inequality is valid for any $\gamma \in \Gamma$, the first inequality follows.

If moreover $\chi_\omega \le h_k$ then $\limsup_{k\to+\infty} \ell^T(h_k) \le \ell^T(\omega) \le \ell^T(h_k)$ and the result follows. □

Note that, in the above proof, we use the fact that $h_k(x) \to \chi_\omega(x)$ for *every* x. Almost everywhere convergence (in the Lebesgue sense) would not be enough.

Remark 4. We denote by d the geodesic distance on (M, g). It is interesting to note that, given any subset ω of M:

- ω is open if and only if there exists a sequence of continuous functions h_k on M satisfying $0 \leq h_k \leq h_{k+1} \leq \chi_\omega$ for every $k \in \mathbb{N}^*$ and converging pointwise to χ_ω.
  Indeed, if ω is open, then one can take for instance $h_k(x) = \min(1, k\, d(x, \omega^c))$. Conversely, since $h_k(x) = 0$ for every $x \in \omega^c$, by continuity of h_k it follows that $h_k(x) = 0$ for every $x \in \overline{\omega^c} = (\mathring{\omega})^c$. Now take $x \in \omega \setminus \mathring{\omega}$. We have $h_k(x) = 0$, and $h_k(x) \to \chi_\omega(x)$, hence $\chi_\omega(x) = 0$ and therefore $x \in \omega^c$. Hence ω is open.
- ω is closed if and only if there exists a sequence of continuous functions h_k on M satisfying $0 \leq \chi_\omega \leq h_{k+1} \leq h_k \leq 1$ for every $k \in \mathbb{N}^*$ and converging pointwise to χ_ω.
  Indeed, if ω is closed, then one can take $h_k(x) = \max(0, 1 - k\, d(x, \omega))$. Conversely, since $h_k(x) = 1$ for every $x \in \omega$, by continuity of h_k it follows that $h_k(x) = 1$ for every $x \in \overline{\omega}$, and thus $\chi_{\overline{\omega}} \leq h_k \leq 1$. Now take $x \in \overline{\omega} \setminus \omega$. We have $h_k(x) = 1$ and $h_k(x) \to \chi_\omega(x)$, hence $\chi_\omega(x) = 1$ and therefore $x \in \omega$. Hence ω is closed.

Lemma 10. *Let ω be an open subset of M and let $T > 0$ be arbitrary. For every sequence of continuous functions h_k on M converging pointwise to χ_ω, satisfying moreover $0 \leq h_k \leq h_{k+1} \leq \chi_\omega$ for every $k \in \mathbb{N}^*$, we have*

$$\ell^T(\omega) = \lim_{k \to +\infty} \ell^T(h_k) = \sup_{k \in \mathbb{N}^*} \ell^T(h_k).$$

Proof. Since $h_k \leq \chi_\omega$, we have $\ell^T(h_k) \leq \ell^T(\omega)$. By continuity of h_k and by compactness of geodesics, there exists a geodesic ray γ_k such that $\ell^T(h_k) = \frac{1}{T}\int_0^T h_k(\gamma_k(t))\, dt$. Again by compactness of geodesics, up to some subsequence γ_k converges to a ray $\bar{\gamma}$ in $C^0([0,T], M)$. We claim that

$$\liminf_{k \to +\infty} h_k(\gamma_k(t)) \geq \chi_\omega(\bar{\gamma}(t)) \qquad \forall t \in [0,T].$$

Indeed, either $\bar{\gamma}(t) \notin \omega$ and then $\chi_\omega(\bar{\gamma}(t)) = 0$ and the inequality is obviously satisfied, or $\bar{\gamma}(t) \in \omega$ and then, using that ω is open, for k large enough we have $\gamma_k(t) \in U$, where $U \subset \omega$ is a compact neighborhood of $\bar{\gamma}(t)$. Since h_k is monotonically nondecreasing and χ_ω is continuous on U, it follows from the Dini theorem that h_k converges uniformly to χ_ω on U, and then we infer that $h_k(\gamma_k(t)) \to 1 = \chi_\omega(\bar{\gamma}(t))$. The claim is proved. Now, we infer from the Fatou lemma that

$$\begin{aligned}\ell^T(\omega) \leq \frac{1}{T}\int_0^T \chi_\omega(\bar{\gamma}(t))\, dt \leq \frac{1}{T}\int_0^T \liminf_{k \to +\infty} h_k(\gamma_k(t))\, dt \\ \leq \liminf_{k \to +\infty} \frac{1}{T}\int_0^T h_k(\gamma_k(t))\, dt = \liminf_{k \to +\infty} \ell^T(h_k) \leq \ell^T(\omega)\end{aligned}$$

and we get the equality. □

Remark 5. The results of Lemmas 9 and 10 are valid as well for subsets of S^*M (which is a metric space).

Lemma 11. *Let ω be an open subset of M and let $T > 0$ be arbitrary. There exists $\gamma \in \Gamma$ such that $\ell^T(\omega) = \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\, dt$, i.e.the infimum in the definition of $\ell^T(\omega)$ is reached.*

Proof. The argument is almost contained in the proof of Lemma 10, but for completeness we give the detail. Let $(\gamma_k)_{k \in \mathbb{N}^*}$ be a sequence of rays such that $\frac{1}{T}\int_0^T \chi_\omega(\gamma_k(t))\, dt \to \ell^T(\omega)$. By compactness of geodesics, $\gamma_k(\cdot)$ converges uniformly to some ray $\gamma(\cdot)$ on $[0,T]$.

Let $t \in [0,T]$ be arbitrary. If $\gamma(t) \in \omega$ then for k large enough we have $\gamma_k(t) \in \omega$, and thus $1 = \chi_\omega(\gamma(t)) \leq \chi_\omega(\gamma_k(t)) = 1$. If $\gamma(t) \in M \setminus \omega$ then $0 = \chi_\omega(\gamma(t)) \leq \chi_\omega(\gamma_k(t))$ for any k. In all cases, we have obtained the inequality

$$\chi_\omega(\gamma(t)) \leq \liminf_{k \to +\infty} \chi_\omega(\gamma_k(t)) \qquad \forall t \in [0,T].$$

By the Fatou lemma, we infer that

$$\begin{aligned}\ell^T(\omega) \leq \frac{1}{T}\int_0^T \chi_\omega(\gamma(t))\, dt \leq \frac{1}{T}\int_0^T \liminf_{k \to +\infty} \chi_\omega(\gamma_k(t))\, dt \\ \leq \liminf_{k \to +\infty} \frac{1}{T}\int_0^T \chi_\omega(\gamma_k(t))\, dt = \ell^T(\omega)\end{aligned}$$

and the equality follows. □

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